NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305214
Lead Applicant	Mr Timothy Jenkins
Research Title	Development, feasibility and acceptability of a diaphragm focussed Inspiratory Muscle Training (IMT) intervention in difficult to wean, mechanically ventilated patients.
Plain English Summary	**Purpose** 200,000 adults are admitted to intensive care in the UK each year. Many of these patients need help with their breathing and are put on a machine called a ventilator. Research shows that many patients using a ventilator quickly develop weakness in their main breathing muscle, called the diaphragm. This weakness can cause a longer time on the ventilator, a longer hospital stay, and higher chances of health problems or death. Inspiratory muscle training is a breathing exercise where the patient breathes against resistance through a small device (like breathing through a straw). It aims to improve the strength of the diaphragm and reduce how long patients are on the ventilator. **Why is this research needed?** Research shows that current breathing exercise programmes do not help patients come off the ventilator or go home quicker. This may be because these breathing exercises strengthen the wrong muscles (those in the chest and neck, that are not efficient). These exercises can also be distressing for patients. This research aims to develop a breathing exercise programme that focusses on strengthening the diaphragm. We will also make sure patients do not find the breathing exercise too distressing. We will then test if it is possible to carry out a large study to see if this programme can help patients come off the ventilator and go home quicker. **What will be done in this research?** Two studies will be done on patients who cannot breathe without a ventilator. **Study 1:** Will measure how the breathing muscles respond to four different levels of breathing exercise and ask patients to feedback on each level. 1. At each breathing exercise level, the effort of the breathing muscles will be measured using a small tube inserted into the nostril and small sticky patches on the chest and neck. We aim to find the level where the diaphragm is being pushed to work hard but other muscles in the chest and neck are not. 2. After each breathing exercise level patients will be asked how they found it – for example hard, easy, uncomfortable, scary, motivating. **The results from both these parts will together help to make a breathing exercise programme which focusses on strengthening the diaphragm and is acceptable to patients.** **Study 2:** Will test if it is possible to carry out a large enough study to see if the breathing exercise programme developed in study 1 works. We will aim to find out if it is possible to recruit and keep patients in the study. Patients will be randomly put into one of two groups by a computer: • One group will perform the breathing exercise programme formed in study 1 (they will perform 60 breaths a day through the device) and usual care. • One group will have usual care. Patients will stay in the study until they can breathe without the ventilator. We will also interview physiotherapists to ask how they felt doing these breathing exercises with patients. **How will patients be involved?** Patients who have spent a long time on a ventilator in intensive care have helped

	design this study. They will continue to be involved when the study is running. They will help design information sheets, consent forms and continue to advise on how to ask patients about their experience of breathing exercises. One person from the group will attend scientific meetings with the researchers. **Telling people about the results of the study** We will tell healthcare professionals about the results at conferences, and by publishing them in journals. We will work with patient advisory groups and charities to inform patients using blogs, newsletters, and social media.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
Guy's and St Thomas' NHS Foundation Trust

Partner organisation	Brunel University

Research Title	Development, feasibility and acceptability of a diaphragm focussed Inspiratory Muscle Training (IMT) intervention in difficult to wean, mechanically ventilated patients.

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2025

Research Type	Feasibility study, Other (please specify)

Please provide more details on your selection above.
Study 1: Mixed methods observational study Study 2: Feasibility Randomised Controlled Trial

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2012	09/2015	BSc(Hons)	Physiotherapy	United Kingdom	Kings College London	First class honours
09/2004	09/2007	BSc(Hons)	Nursing Studies	United Kingdom	Kings College London	2:1

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
01/2024	-	NIHR ICA PCAF Research Physiotherapist - Critical Care and Transplantation	Harefield Hospital, Guy's and St Thomas' NHS Foundation Trust, London	100
09/2023	-	Editorial Board Member - Association of Chartered Physiotherapists in Respiratory Care (ACPRC)	Association of Chartered Physiotherapists in Respiratory Care (ACPRC)	100
02/2023	01/2024	Research Physiotherapist - Harefield Respiratory Research Group	Guy's and St Thomas' NHS Foundation Trust, London	100
02/2023	-	Honorary Research Fellow	Brunel University, London	100
07/2022	02/2023	Highly Specialist Physiotherapist in Transplantation and Critical Care	Harefield Hospital, Guy's and St Thomas' NHS Foundation Trust, London	0
09/2021	07/2022	Research Fellow - The Royal Brompton and Harefield Hospital Charity Research Fellowship	Royal Brompton and Harefield Hospitals, Guy's and St Thomas' NHS Foundation Trust, London	100
10/2018	09/2021	Band 6 Rotational Physiotherapist	Royal Brompton & Harefield NHS Foundation Trust	0
01/2016	10/2018	Band 5 Rotational Physiotherapist	Chelsea and Westminster NHS Foundation Trust	0
09/2015	01/2016	Band 5 Rotational Physiotherapist	Barts Health NHS Foundation Trust	0
09/2007	08/2012	Band 5 Nurse - General Intensive Care	St George's University Hospitals NHS Foundation Trust	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Royal Brompton and	Grant for: **'Measurement of the Metabolic Cost of Inspiratory Muscle training and Physical**	43175	The Royal Brompton and Harefield	10	10/09/2021	10/07/2022

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Harefield Hospitals Charity Fellowships 2020, round 2	**Rehabilitation in mechanically ventilated patients; a feasibility study'** ClinicalTrials.gov number: NCT05101850. Involvement: Project design, management, day to day running, recruitment, data collection, analysis, manuscript preparation. Publication DOI: https://doi.org/10.1186/s40635-023-00522-6 Physical rehabilitation manuscript estimated publication date: Sept 2024		Hospitals Charity			
Brunel Partners Academic Centre for Health Sciences	Co-applicant with Dr Vicky MacBean to conduct **'The metabolic cost of inspiratory muscle training in healthy volunteers'** at Brunel University, London. Involvement: Project design, MSc student supervision, analysis, abstract and manuscript preparation. Two abstracts presented at The Physiological Society conference, 2023. Full manuscript under preparation, estimated publication date: November 2024	1000	Brunel Partners Academic Centre for Health Science	3	01/06/2022	01/09/2022
National Institute for Health Research ICA Pre-doctoral Clinical and Practitioner Academic Fellowship (PCAF)	Main applicant. PPI engagement; clinical exercise, qualitative and quantitative research methods MSc modules; statistical and coding training; PPI training. Healthy inspiratory muscle training study at Brunel University measuring diaphragm and extra-diaphragmatic muscle activity Involvement: study design, ethics, recruitment, study procedures, analysis and future manuscript preparation Estimated publication date: March 2025	73953	National Institute for Health Research (NIHR)	12	01/01/2024	31/12/2024

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Jenkins, T., Patel, S., Edwards, G., Nolan, C., Canavan, J., Kon, S., Jones, S., Barker, R., Littlemore, Maddocks, M. and Man, W.. (2024) ‘Longitudinal change in ultrasound-derived rectus femoris cross-sectional area in COPD’, *ERJ open research*, (in press).
Edwards, G., Patel, S., **Jenkins, T.**, Ingram, K., Man, W. and Nolan, C.. (2023) ‘Estimation of the minimal important difference (MID) for the 1 repetition max (1RM) in COPD’, *European Respiratory Journal 62 (suppl 67)* .
Trivedi, P., Patel, S., Edwards, G., **Jenkins, T.**, Man, WD. and Nolan, CM.. (2023) ‘Five-Repetition Sit-to-Stand Test: Responsiveness and Minimal Important Difference in Idiopathic Pulmonary Fibrosis.’, *Annals of the American Thoracic Society*.
Scott, W., Aristidou, A., Polkey, MI., **Jenkins, T.** and MacBean, V.. (2023) ‘Influence of anxiety, breathing vigilance and respiratory muscle strength on respiratory load perception’, *Proc Physiol Soc* , (54, PCB053).
Aristidou, A., Scott, W., Polkey, MI., **Jenkins, T.** and MacBean, V.. (2023) ‘The metabolic cost of inspiratory muscle training in healthy adults’, *Proc Physiol Soc* , (54, PCA052).
Jenkins, TO., MacBean, V., Poulsen, MK., Karbing, DS., Rees, SE., Patel, BV. and Polkey, MI.. (2023) ‘The metabolic cost of inspiratory muscle training in mechanically ventilated patients in critical care.’, *Intensive care medicine experimental*, 11(1).
Jenkins, T., Poulsen, MK., Karbing, DS., Rees, SE., Patel, B., Polkey, MI. and MacBean, V.. (2023) ‘The metabolic cost of inspiratory muscle training in mechanically ventilated patients.’, *American journal of respiratory and critical care medicine*, 2023;207:A2882.
Patel, S., Harvey, J., **Jenkins, T.**, Nolan, C. and Man, W.. (2023) ‘Screening for Clinically Relevant Thresholds of the Incremental Shuttle Walk in People With COPD Using the One-minute Sit to Stand’, *American journal of respiratory and critical care medicine*, (207:A1025).
Man, W., Littlemore, Patel, S., Harvey, J., Ingram, K., **Jenkins, T.** and Nolan, C.. (2023) ‘Video Telerehabilitation Versus Centre-based Pulmonary Rehabilitation: A Real-World Propensity Matched Analysis’, *American journal of respiratory and critical care medicine*, (207:A1027).
Jenkins, T., Patel, S., Canavan, J., Kon, S., Jones, S., Barker, R., Littlemore, Harvey, J., Nolan, C. and Man, W.. (2022) ‘Longitudinal change in Ultrasound derived Rectus Femoris cross sectional area (RFCSA) in patients with COPD’, *Eur Respir J 2022; 60: Suppl. 66, 764.*.
Harvey, JH., Patel, S., Littlemore, **Jenkins, T.**, Nolan, C. and Man, W.. (2022) ‘Validation of the one-minute sit to stand test (1MSTS) for identification of exertional oxygen desaturation in chronic respiratory disease.’, *Eur Respir J 2022; 60: Suppl. 66, 1454*.
Jenkins, T., Beverley, Z. and Jones, A.. (2020) ‘Yoga as exercise for cystic fibrosis inpatients’, *Journal of Cystic Fibrosis*, (2020; Suppl. 19S2).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/11/2023	Royal Brompton and	Directors Choice Award:	First author on abstract

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
	Harefield Hospitals	The annual Royal Brompton and Harefield Hospitals research showcase for the Metabolic Cost of Inspiratory Muscle Training in Mechanically Ventilated Patients in Critical Care	and paper
01/01/2023	The Council for Allied Health Professions Research (CAHPR) London with UCL AHPs in Academia Research	£50 prize for best oral presentation for 'The Metabolic Cost of Inspiratory Muscle Training in Mechanically Ventilated Patients'	Presenter
01/06/2022	CAHPR London Hub conference bursary funding	£250 to attend the European Respiratory Society (ERS) international conference to present 'Longitudinal change in Ultrasound derived Rectus Femoris cross sectional area (RFCSA) in patients with COPD'	Poster presentation at ERS
01/12/2021	Rehabilitation after Critical Illness. Brunel University MSc module	Awarded 1st in 3000-word assignment	Advanced clinical practice student
01/09/2019	Royal Brompton Physiotherapy poster event	Awarded best change to a process for 'Does a Physiotherapy led formal education and exercise intervention improve attitudes, behavior's and exercise capacity in paediatric CF patients?'	Lead on project, presenter

ORCID iD 0000-0002-8631-0725

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
PH107591 - HCPC

Please describe your research career to date
Since 2021, I have secured a total of £118,128 in personal research grants and been lead investigator for three clinical and three lab-based studies. **2021-2022** In 2021 I was awarded the competitive Royal Brompton and Harefield Hospitals (RBHT) Charity research fellowship (£43,175). In collaboration with Dr MacBean, Professor Polkey and experts at Aalborg University, Denmark we designed a novel observational study to understand the physiological response to rehabilitation and Inspiratory Muscle Training (IMT) in mechanically ventilated patients in intensive care (ICU). IMT is an exercise aimed at improving breathing muscle strength to help critically ill patients breathe without a ventilator. In leading this seven-month study, I doubled the original recruitment target from 20 to 40, conducting 106 measurements. I was responsible for all day-to-day management including recruitment, data analysis and manuscript preparation. The study involved a challenging three-hour exercise protocol in mechanically ventilated patients, collecting specialised physiological data. The results described which patients might better tolerate IMT in ICU. I presented these data at the American Thoracic Society international conference, 2023. The results are now published (first author) in *Intensive Care Medicine Experimental* (IF 3.5). I was awarded “Directors’ Choice” prize in the annual RBHT research showcase for this work. A further manuscript reporting secondary outcomes is under peer review. To broaden my research skills, I conducted two additional lab studies within the fellowship to build on results of the ICU study. I organised a one-week visit to Aalborg University, where I worked with academics to conduct a robust IMT research protocol. We measured oxygen consumption during four different IMT loads in ten participants. After completing a coding course, I will analyse these data later in my PCAF during a second visit. Brunel Partners Academic Centre for Health Sciences funded a second healthy study (£1000) to measure load perception in addition to oxygen consumption during IMT. Under the supervision of Dr MacBean, I supported two MSc students in study techniques and data collection at Brunel University, where we recruited 30 participants in two months. I analysed the data in collaboration with Aalborg University. I supported the MSc students to present two abstracts at The Physiological Society conference, 2023. I am first author on the manuscript in preparation. **2023-2024** In 2023 I moved to the Harefield Respiratory Research Group to enhance my research skills and knowledge in

larger, NIHR-funded trials. I was the lead investigator in a trial investigating a novel app for patients attending pulmonary rehabilitation. I was responsible for all day-to-day running of the trial, where I recruited 69 patients (target: 50) in four months. I led the data analysis and am first author for the manuscript in preparation. I wrote up data previously presented at the European Respiratory Society congress (funded by a Council for Allied Health Professions Research conference award), now published as a full article in *ERJ Open Research* (IF 4.6).

In addition to my full-time role, I was the principal investigator on an observational study investigating a modification to reduce oxygen consumption in ventilators, in collaboration with clinical engineering and Professor Polkey. I was responsible for all aspects of the trial, including ethics, recruitment, and a complex two-hour study procedure in thirteen mechanically ventilated patients. I led the analysis and manuscript preparation (currently under peer review).

2024

The NIHR PCAF has enabled extensive PPI activity, PPI training (Public Involvement in Research, Imperial College London (ICL)), and to conduct a lab study to inform specific elements of my DCAF. Under the supervision of Dr MacBean, I learnt techniques in respiratory electromyography (EMG). I conducted a complex EMG study in healthy participants to refine the protocol used in study 1 of this DCAF.

I attended MSc modules in qualitative and quantitative research methods (Brunel University) and in clinical exercise (UCL). I will attend a five-day researcher development course at UCL, and a five-day statistics for medical research course at ICL later in my PCAF. These formal courses have increased my broader understanding of research design, statistics and methodology which lead well into the programme of work in my DCAF. I convened a PPI group with which I have actively engaged throughout planning this DCAF.

Professional experience

I am a highly specialist respiratory physiotherapist with an interest in physical rehabilitation, weaning and respiratory rehabilitation after critical illness. I have over six years' experience working at the RBHT specialising in critical care and surgery, building on extensive knowledge derived from five years as an ICU nurse.

I realise development as a clinical academic involves more than just conducting research. Clinically, I regularly teach physiotherapists and medics in critical care. I am an honorary research fellow at Brunel University and lecture on two critical care advanced clinical practice MSc modules. I attended the Starting Out in Research course at ICL in 2020, and was subsequently asked to mentor on the course, a role I continue.

Under the supervision of Professor McWilliams, I am forging a clinical academic role at Harefield Hospital, where I have championed research by supporting colleagues with audits, service evaluations and quality improvement projects. These will lead to our work being presented at Trust research events and submitted to conferences.

As an editorial board member at the Association of Chartered Physiotherapists in Respiratory Care, I am leading a team of 11 to conduct an IMT scoping review of over 2900 articles, to be published next year. This role has provided an up-to-date review of the IMT literature and supported my development through leadership and networking.

Future aspirations

I aspire to be a leading clinical academic with an international profile in weaning and rehabilitation in critical care. The training, practice supervision and skills I develop during this DCAF will enable me to conduct further research and embed a research culture into clinical practice to improve patient care. Within one year of completion, I would apply for postdoctoral funding to conduct a definitive RCT from the results of this DCAF. The strong track record for supporting clinical academic careers at RBHT/GSTT mean I am well positioned for this progression.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors

Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

4. Plain English Summary of Research

Purpose

200,000 adults are admitted to intensive care in the UK each year. Many of these patients need help with their breathing and are put on a machine called a ventilator.

Research shows that many patients using a ventilator quickly develop weakness in their main breathing muscle, called the diaphragm. This weakness can cause a longer time on the ventilator, a longer hospital stay, and higher chances of health problems or death.

Inspiratory muscle training is a breathing exercise where the patient breathes against resistance through a small device (like breathing through a straw). It aims to improve the strength of the diaphragm and reduce how long patients are on the ventilator.

Why is this research needed?

Research shows that current breathing exercise programmes do not help patients come off the ventilator or go home quicker. This may be because these breathing exercises strengthen the wrong muscles (those in the chest and neck, that are not efficient). These exercises can also be distressing for patients.

This research aims to develop a breathing exercise programme that focusses on strengthening the diaphragm. We will also make sure patients do not find the breathing exercise too distressing. We will then test if it is possible to carry out a large study to see if this programme can help patients come off the ventilator and go home quicker.

What will be done in this research?

Two studies will be done on patients who cannot breathe without a ventilator.

Study 1:

Will measure how the breathing muscles respond to four different levels of breathing exercise and ask patients to feedback on each level.

1. At each breathing exercise level, the effort of the breathing muscles will be measured using a small tube inserted into the nostril and small sticky patches on the chest and neck. We aim to find the level where the diaphragm is being pushed to work hard but other muscles in the chest and neck are not.
2. After each breathing exercise level patients will be asked how they found it – for example hard, easy, uncomfortable, scary, motivating.

The results from both these parts will together help to make a breathing exercise programme which focusses on strengthening the diaphragm and is acceptable to patients.

Study 2:

Will test if it is possible to carry out a large enough study to see if the breathing exercise programme developed in study 1 works. We will aim to find out if it is possible to recruit and keep patients in the study.

Patients will be randomly put into one of two groups by a computer:

- One group will perform the breathing exercise programme formed in study 1 (they will perform 60 breaths a day through the device) and usual care.
- One group will have usual care.

Patients will stay in the study until they can breathe without the ventilator.

We will also interview physiotherapists to ask how they felt doing these breathing exercises with patients.

How will patients be involved?

Patients who have spent a long time on a ventilator in intensive care have helped design this study. They will continue to be involved when the study is running. They will help design information sheets, consent forms and continue to advise on how to ask patients about their experience of breathing exercises. One person from the group will attend scientific meetings with the researchers.

Telling people about the results of the study

We will tell healthcare professionals about the results at conferences, and by publishing them in journals. We will work with patient advisory groups and charities to inform patients using blogs, newsletters, and social media.

5. Scientific Abstract

Background

Diaphragm weakness is present in 63-80% of patients mechanically ventilated for over 48 hours and is associated with increased length of intensive care and hospital stay, greater cognitive and functional impairments and increased mortality risk.

The diaphragm is responsible for 80% of work in normal breathing, therefore specific diaphragm strengthening may help patients wean from mechanical ventilation. 63% of intensive care physiotherapists worldwide use Inspiratory Muscle Training (IMT), which aims to improve diaphragm strength through the application of resistance during inspiration, using a device connected directly to the patient's tracheostomy. However, there are a number of issues with current IMT approaches in intensive care. There is no consensus on how to specifically train the diaphragm. High-intensity IMT will substantially recruit extra-diaphragmatic inspiratory muscles and type IIb (power) diaphragm muscle fibres; neither are useful for ventilator weaning. Current approaches to IMT in mechanically ventilated patients do not improve clinical outcomes, and can be unpleasant and distressing for patients.

Research question

Can a **D**iaphragm-focussed and patient-acceptable **I**nspiratory **M**uscle **T**raining load (D-IMT) be identified and tested in difficult-to-wean, mechanically ventilated patients?

Aims

1. Develop a diaphragm-focussed and patient-acceptable IMT load (D-IMT).

2. Test the feasibility of conducting a randomised controlled trial comparing the D-IMT load with usual care.

Objectives

Study 1

1. Measure diaphragmatic and extra-diaphragmatic inspiratory muscle activity during four IMT loads.

2. Explore the acceptability of these different IMT loads using qualitative methods in line with participants' communication ability.

3. Integrate quantitative and qualitative data to determine a diaphragm-focussed and patient acceptable IMT load (D-IMT)

Study 2

Conduct a feasibility randomised controlled trial (RCT) to ascertain if it is possible to conduct a definitive RCT to test the clinical- and cost-effectiveness of D-IMT compared to usual care.

Methods and timelines

Study 1 (months 1-17).

An observational, mixed methods study. Forty difficult-to-wean, mechanically ventilated participants will perform ten breaths of IMT at 15, 25, 35 and 45% of their maximal inspiratory strength on up to three occasions. During

IMT, diaphragm activity will be measured using an oesophageal electromyography (EMG) catheter. Extra-diaphragmatic inspiratory muscle activity will be measured using surface EMG. Participant experience of each load will be explored qualitatively. These data will be integrated to inform a diaphragm-focussed, patient-acceptable IMT load (D-IMT).

Study 2 (months 18-36). Feasibility RCT. Fifty difficult-to-wean, mechanically ventilated participants will be randomised to usual care or D-IMT and usual care. The D-IMT group will perform the IMT load defined in study 1 five days per week until they no longer require mechanical ventilation. Feasibility outcomes: eligible patients, recruitment rates, completeness of outcome measures, IMT fidelity, patient adherence to IMT, participant retention, adverse events. Semi-structured interviews will be conducted on physiotherapists who administered D-IMT to explore acceptability of the intervention.

Anticipated impact and dissemination

Findings will be disseminated via peer-reviewed publications, conference presentations and patient and public events in collaboration with the project's PPI group. Results of study 2 will inform a definitive RCT with the potential to improve clinical and patient-related outcomes and reduce NHS costs.

6. Detailed Research Plan

What is the problem being addressed?

Diaphragm dysfunction in mechanically ventilated patients

Around 200,000 adult patients are admitted to intensive care (ICU) in the UK each year[1], of whom 50% require invasive mechanical ventilation[2]. Diaphragmatic dysfunction (weakness) is present in 63-80% of patients mechanically ventilated for over 48 hours, and is associated with prolonged ventilation, increased mortality, and readmission to ICU[3-8]. Patients who require prolonged ventilation ('difficult to wean' patients) have increased long-term cognitive and functional impairments, and poorer post-discharge survival[7,9]. Difficult to wean patients occupy a disproportionate number of ICU bed days, with increased healthcare costs[10,11].

Inspiratory muscle training

Inspiratory Muscle Training (IMT) has been proposed as treatment strategy for difficult to wean patients[12]. IMT aims to improve the strength and endurance of the diaphragm through the application of resistance during inspiration[13,14], using a small device which is connected directly to the patient's tracheostomy.

However, although IMT has been shown to improve overall volitional inspiratory muscle strength (of the diaphragm *and* the extra-diaphragmatic muscles), two systematic reviews have concluded there is insufficient evidence to confirm if it has a positive effect on clinical outcomes such as weaning duration, and ICU and hospital length of stay[14,15] There is currently no consensus on how to specifically train the diaphragm adequately in mechanically ventilated patients[16], and significant heterogeneity in training protocols reported in the literature, with no clear physiological basis for training regimes chosen[16-23].

The diaphragm is the primary respiratory muscle[24], yet frequently cited IMT loads will recruit non-diaphragmatic inspiratory muscles, not useful for sustained respiratory effort in mechanically ventilated patients [16,25,26]. Additionally, mechanically ventilated patients experience moderate to severe breathlessness during loaded breathing in ICU[27]; a sensation consistently associated with the fear of dying[28]. IMT is unpleasant and distressing for patients; members of the PPI group for this project describe high load IMT in ICU "*like dying*" or "it *was an extremely hard task".* One member quoted that *"recollections of doing IMT formed hallucinations, dreams and nightmares which I experienced during and after my ICU stay"*. There are calls for the optimisation of the IMT stimulus in difficult to wean patients[,29,30] using the principles of muscle physiology and exercise prescription[16].

This project will address these problems by:

1. Developing a diaphragm focussed and patient acceptable IMT regime (D-IMT) using an embedded mixed method approach.
2. A randomised feasibility study will determine whether it is possible to conduct a larger randomised controlled trial (RCT) implementing D-IMT compared to usual care.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to service users and health and social care services?

Consequences of diaphragm weakness

Diaphragm weakness is a marker of illness severity and poor prognosis in mechanically ventilated patients[3,31,32]. Prolonged ventilation (associated with diaphragm weakness) impacts patients significantly, and is associated with disability at six months post-ICU discharge, activity limitation including return to work, reduced quality of life, and cognitive impairments[7,9,33,34].

The patient experience of IMT

The most commonly used IMT regimes[13] are unpleasant for patients, who can experience severe breathlessness during and after IMT in ICU. Breathlessness is a major stressor for mechanically ventilated patients[35], which can contribute to post-traumatic stress syndrome in ICU survivors[36].

Healthcare resources

The optimal IMT load to use in clinical practice which improves clinical outcomes is yet to be defined[14,29,30]. However, 63% of physiotherapists worldwide administer IMT in ICU[37]. A survey I conducted through Brunel University found 50% of UK physiotherapists use IMT in ICU, meaning NHS resources are being used to deliver an intervention lacking a robust evidence base.

These data show the importance of this research. Finding the optimal approach to IMT may:

1. Help patients wean from mechanical ventilation quicker. This could ameliorate the negative effects of prolonged ventilation to improve outcomes, and reduce NHS costs through a reduction in ICU and hospital bed days.
2. Improve patient wellbeing in ICU by reducing stressors associated with IMT.
3. Improve resource use by informing clinical practice with robust evidence.

Review of existing evidence

Seven RCTs have investigated IMT in mechanically ventilated patients **(Table 1).** There is substantial heterogeneity in loads used, set anywhere between 20 and 50% of the patient's inspiratory strength, then increased to a load where the patient can just complete the sixth breath of IMT[17-23]

Table 1: Details of RCTs investigating IMT in mechanically ventilated patients

Author (year)	IMT intervention	Outcomes measured	Outcome Improved with IMT?
Cader et al. (2010)	Low load endurance	PImax	Yes
		Weaning duration	No
Martin et al. (2011)	High load	PImax	Yes
		Weaning success	Yes
Condessa et al. (2013)	Moderate load	PImax	Yes
		Weaning duration	No
Moreno et al. (2019)	High load	PImax	No
		Weaning duration	No
Bissett et al. (2022)	High load	PImax	No
		Breathlessness at rest	Yes
		Weaning duration	No
Ratti et al. (2022)	Moderate load	PImax	Yes
		Weaning duration	No
Van Hollebeke et al. (2024)	High load	PImax	No
		Weaning duration	No

Two systematic reviews concluded that while IMT may improve volitional measures of overall inspiratory muscle strength, this does not translate to improvements in meaningful clinical outcomes such as weaning duration or success[14,15].

Justification for a physiologically informed, acceptable load

The diaphragm is the primary respiratory muscle, responsible for around 80% of respiratory work, recruiting mainly type I and IIa endurance fibres in normal breathing in healthy adults[16,24]. Importantly, it is *specific diaphragm weakness* that is strongly associated with weaning failure in mechanically ventilated patients, not overall inspiratory strength (encompassing the diaphragm and extra-diaphragmatic inspiratory muscles)[3,32,38]. IMT will not only target the diaphragm, but to a larger extent the extra-diaphragmatic inspiratory muscles as the load increases[39]. High load IMT is frequently used in difficult to wean patients[18,20,21,23].

During my PCAF, under the supervision of Dr MacBean we collected novel data during IMT at eight loads in healthy subjects. As IMT load increases, there is a proportionally greater increase in extra-diaphragmatic muscle activation compared to diaphragmatic activation at higher IMT loads.

High load IMT causes a training effect on the scalene and sternocleidomastoid muscles (extra-diaphragmatic muscles), but does not improve weaning success[29], questioning the utility of high load IMT. Although the extra-diaphragmatic muscles are a mechanism of compensation that can be activated in the presence of increased respiratory load (for example, during exercise), they are not useful for sustained ventilatory effort in mechanically ventilated patients[25,26]. Greater activation of extra-diaphragmatic inspiratory muscles and concurrent diaphragmatic dysfunction are associated with an increased likelihood of failing a ventilator weaning trial[26,40,41].

These data emphasise the importance of limiting extra-diaphragmatic muscle activation breathing in critically ill patients and build a case for targeted diaphragmatic IMT in difficult to wean patients.

The importance of patient experience and feedback in defining the optimal IMT load cannot be underestimated and is why this study uses a mixed methods approach. IMT involves the physiotherapist connecting the IMT device directly to the patient's tracheostomy. The patient is compelled to breathe through the applied resistance to obtain tidal volume. An acceptable load would avoid distressing experiences for patients as previously reported[27,35], and as highlighted by the project PPI group. It may also reduce IMT refusal rates, potentially improving the effectiveness of the intervention. Percentage of intended IMT sessions completed varies from 30 to 88% in the literature[18,21]. Refusal or fatigue are frequent reasons for not completing IMT[21], which may be related to the unpleasant nature of IMT.

Research question

Can a **D**iaphragm-focussed and patient-acceptable **I**nspiratory **M**uscle **T**raining load (D-IMT) be identified and tested in difficult to wean, mechanically ventilated patients?

Aims

1. Develop a diaphragm-focussed and patient-acceptable IMT load (D-IMT).

2. Test the feasibility of conducting a randomised controlled trial (RCT) comparing the D-IMT load with usual care.

Objectives

Study 1

1. Measure diaphragmatic and extra-diaphragmatic (scalene, sternocleidomastoid, and pectoral) inspiratory muscle activation using electromyography (EMG) during four IMT loads in difficult to wean patients.

2. Explore the acceptability of these different IMT loads using intensity and unpleasantness numeric rating scales and qualitative methods in line with their communication needs.

3. Integrate quantitative and qualitative data to determine a diaphragm-focussed and patient acceptable IMT load (D-IMT).

Study 2

Conduct a feasibility RCT to determine if it is possible to conduct a definitive RCT to test the clinical- and cost-effectiveness of D-IMT compared to usual care. Results will guide the design and powering of a definitive RCT, aligning with the Medical Research Council guidance on complex interventions[42].

Project plan

The research support service and a statistician were consulted in the design of this plan.

Study 1: A prospective, mixed methods, observational study to define a diaphragm-focussed and patient-acceptable IMT load (D-IMT) in difficult to wean, mechanically ventilated patients.

(Months 1-17)

Design

The complexity of IMT in mechanically ventilated patients who have compromised means to communicate warrants an embedded mixed methods approach; Neither a purely qualitative nor quantitative approach would gain sufficient understanding of the problem. In this design quantitative and qualitative data will be collected simultaneously, embedding qualitative data within quantitative data. This approach allows the focus to be on the quantitative data necessary to determine the parameters for diaphragm-focussed IMT whilst the qualitative data will provide crucial insights into patients' experiences[43].

Participants will perform four IMT loads, where diaphragmatic and extra-diaphragmatic muscle activation will be measured. After each of the four IMT loads, and the entire session, qualitative data will be collected. It is recognised that, due to delirium or impaired cognition, there will be variability in participants' ability to engage in qualitative inquiry about their experience. The PPI group felt strongly that data collection over 1-3 occasions would improve data quality, which will also account for changes in condition, cognition and weaning status, as previously described[44]. Patients meeting eligibility criteria will not be excluded due to confusion or impaired cognition at consent.

Figure 1 details the design of study 1.

Inclusion criteria

- Mechanically ventilated patients in whom weaning has not been achieved within 24 hours after a first weaning attempt[7]
- Aged ≥ 18 years
- Co-operative and able to participate in IMT.
- Informed consent or surrogate approval.

Exclusion criteria

- Facial or skull fracture, or a bleeding disorder which would prohibit oesophageal catheter placement.

- Undrained pneumothorax/pneumomediastinum.
- Pregnancy
- Unlikely to survive, or consultant assessment that the patient is not appropriate.
- Patient requiring internal or external cardiac pacing.
- Wounds that prohibit EMG electrode placement.

Consent

By preference, informed consent will be sought. However, some patients will be able to participate in IMT but not fully understand the study to give informed consent, due to delirium or sedation. In such cases, surrogate consent will be sought from a personal consultee, and retrospective consent sought when capacity is regained. These arrangements are how I conducted my previous IMT study in ICU[45] and will be subject to ethical approval.

Sample size

This is the first study of its kind to measure inspiratory muscle activation and explore patient experience during different IMT loads in mechanically ventilated patients. No formal sample size calculation has been performed; however, an RSS statistician has advised. Based on data from the healthy study in my PCAF, and accounting for the anticipated heterogeneity in the ICU population, we hypothesise that studying 40 participants on 1-3 occasions will allow adequate precision. This target is feasible in a 12-month recruitment window; during my charity fellowship I recruited 40 mechanically ventilated patients with similar eligibility criteria, conducting 106 measurements in geographically different locations over seven-months.

Data collection (Months 2-13)

Quantitative data collection methods

Inspiratory muscle training

IMT will be performed using a PowerBreathe KH2 inspiratory muscle trainer. The device delivers a specific resistance during inspiration and is safe in mechanically ventilated patients[29,46]. Participants will be disconnected from the ventilator, connected to the IMT device, and asked to inhale against the resistance 10 times. This procedure of disconnecting patients from ventilation is cited in all ICU IMT literature, with few adverse events reported, and all non-serious[14,29,46].

Participants will perform IMT at 15, 25, 35 and 45% of their maximal inspiratory pressure in random order, with 20 minutes between loads to allow for recovery and qualitative data collection. The IMT loads arise directly from the study conducted during my PCAF, where we tested eight IMT loads in healthy individuals. We noted a proportional increase in diaphragm activation with increasing IMT load, which, from 30-40% of maximal inspiratory strength was accompanied by a proportionally greater increase in extra-diaphragmatic muscle activity. Testing around this range will allow us to identify a load which maximises diaphragm activation up to the highest level achievable, but below the upward inflection point of extra-diaphragmatic muscle activity.

Measurement of inspiratory muscle activation during IMT

Diaphragm activation will be measured using an oesophageal multipair EMG catheter (NAVA, Getinge, Sweden) designed for mechanically ventilated patients. The catheter also acts as a nasogastric feeding tube; therefore, any existing nasogastric tube will be removed with agreement of the clinical team. Activation of extra-diaphragmatic inspiratory muscles (scalene, sternocleidomastoid, and pectoral) will be measured using surface EMG. I learnt the required skills for oesophageal catheter placement, EMG data collection, and EMG analysis during my PCAF. Diaphragm and extra-diaphragmatic EMG is feasible[47-49] and relates to effort in mechanically ventilated patients[47].

Perceived difficultly and unpleasantness.

After each IMT load, perceived breathing difficulty and unpleasantness will be measured using a 0-10 numeric rating scale, as previously described in ICU[28] with anchor points informed by the project's PPI group: "0=none", "10=extreme".

Qualitative data collection methods

Participant experience will be explored after each IMT load. Recognising the challenges of communicating with participants in ICU, extensive PPI engagement and collaboration with the ICU lead speech and language therapist has informed this element of this study.

Participants will have a spectrum of communication abilities, therefore a range of methods will be employed, developed with the ICU lead speech and language therapist and informed by evidence[44,50,51]:

- Inline speaking valve
- Above cuff vocalisation
- Bespoke communication boards
- Written communication
- Lip-reading
- Electrolarynx

Participants will have varying degrees of abilities to impart anything from basic to complex information[50], influenced by weakness, cognition, delirium, fatigue and sedation[52]. For this reason, a generic qualitative inquiry approach will be used to respond to the different needs and abilities of participants. The generic qualitative approach aims at understanding how individuals make meaning of a phenomenon or situation. In this pragmatic approach, a 'toolbox' of data collection methods will be used. Extensive and ongoing discussions with the project's PPI group have helped inform the content and method of data collection, including:

1. Semi-structured interviews
2. Informal interviews
3. Picture and word boards depicting words used by the PPI group to describe IMT, such as *"Fear, apprehension, anxiety, fatigue, extremely difficult, chest tight, very tired, confidence, positive, effort, pleasant".*

Semi-structured and informal interviews been successfully performed previously in mechanically ventilated and recently weaned patients in ICU[44]. Qualitative data will be recorded using a voice recorder. Fieldwork notes detailing contextual factors and patient observation will provide supplementary data to improve the richness of the qualitative data.

Data analysis (Months 14-15)

Quantitative

Mean peak per-breath activity EMG of inspiratory muscle activation will be presented as percentage contribution of the diaphragm versus extra-diaphragmatic inspiratory muscles. Data will be tested for normality and presented accordingly. Percentage activation for each IMT load will be compared using a one-way ANOVA or Kruskal-Wallis test.

The aim is to identify the IMT load with the maximum diaphragmatic contribution to breathing and an acceptable perceived level of unpleasantness. This will be achieved through visual inspection of individual plots of percentage diaphragmatic activity against load, as per my PCAF study. "Acceptable" unpleasantness will be informed by the qualitative data.

Qualitative

Audio-recordings of qualitative data will be transcribed verbatim (using a professional service in line with GDPR), anonymised and managed through NVivo software V12.6.1 and analysed using the six steps of Reflexive Thematic Analysis[53]. This involves coding at sentence level and developing categories and overarching themes while remaining firmly wedded to the data and context. My supervisor (CK) will independently code one transcript for comparison and validation of themes. Fieldnotes will be drawn upon to provide additional contextual information during the analysis process, this will be particularly important for participants with limited ability to communicate beyond a few words at a time.

Integration of quantitative and qualitative data

Following the separate analysis of the quantitative and qualitative data sets, the data will be integrated and presented in side-by-side joint displays (agreement, partial agreement and silence) to ascertain new insights beyond the information obtained from the separate quantitative and qualitative results[54,55].

Study 2: A feasibility RCT implementing a diaphragm-focussed, patient-acceptable IMT load (D-IMT) compared to usual care

(Months 18-36)

Design

A single centre randomised controlled feasibility study **(Figure 1).** Difficult to wean, mechanically ventilated patients will be randomised to receive either the D-IMT load (informed by study 1) or usual care. Participants will remain in the study until they are successfully weaned from mechanical ventilation.

D-IMT group: Participants will receive usual care and perform IMT 5 days a week: 6 sets of 10 breaths at the load identified in study 1, administered by a costed physiotherapists to ensure adequate blinding for secondary outcome measures.

Control group: Usual care

Randomisation

Participants will be randomised 1:1 to D-IMT or usual care after consent, using a web-based randomisation system provided by REDCap, by a costed physiotherapist for allocation concealment.

Data collection (Months 18-32)

Feasibility outcomes:

- Number of eligible patients
- Recruitment (willingness of eligible patients to consent, reason for non-enrolment)
- Rate of clinician refusal
- Protocol compliance (completeness of secondary outcome measures)
- Intervention fidelity (participant adherence to D-IMT; percentage of intended breaths, sessions and load performed – logged on the PowerBreathe IMT device)
- Study retention (percentage of withdrawals over study period)
- Adverse events in both groups

Secondary (clinical) outcome measures

Patient-centred outcomes

Patient-centred outcomes require greater attention in trials investigating IMT in mechanically ventilated patients[14]. Remarkably, just one IMT study has measured patient-centred outcomes in mechanically ventilated patients[21]. Breathlessness is the second most recalled traumatic experience in ICU survivors[35], highlighting its importance as an outcome measure. Extensive work with the project's PPI group identified weaning time, ICU and hospital length of stay, quality of life, breathlessness, and physical function as important secondary outcome measures.

- **Weaning duration** Defined as time from initiation of mechanical ventilation to successful weaning of mechanical ventilation (48 hours of unassisted breathing[32])
- **ICU and hospital length of stay**
- **Quality of life.** Measured one-week after successful weaning using the SF-36 and EQ-5D-3L. The SF-36 is reliable, is responsive, and has construct and criterion validity in ICU survivors[56]. The EQ-5D-3L tool has been used extensively in ICU survivors[57].
- **Breathlessness**. Measured on enrolment and after successful weaning using a 0-10 numeric rating scale, as previously described[28] with anchor points informed by the project's PPI group: "0=none", "10=extreme". Participants will be asked to report their average and peak breathlessness experienced in the last 24h[21].
- **Function.** Measured one-week after successful weaning using the Chelsea critical care physical assessment tool (CPAx), which measures respiratory function, functional ability and grip strength. It has construct and cross-sectional validity across the ICU and hospital stay, with low floor and ceiling effects[58].
- **Diaphragm function measured by ultrasound** (every three days). Although volitional maximal inspiratory pressure (PImax) is frequently used as a primary outcome measure in IMT studies, and may improve after IMT[14,15,17-19,22] it is neither robust, clinically-relevant nor patient-centred. PImax has similarities with the manoeuvres used during IMT, therefore improvements can be the result of a learning effect of the intervention, rather than a true improvement in diaphragm strength[59]. Furthermore, only a weak correlation exists between PImax and non-volitional measures of specific diaphragm strength[60]. A diaphragm thickening fraction ((end-inspiratory thickness − end-expiratory thickness)/end-expiratory thickness)) <29% detects diaphragm dysfunction (sensitivity 85%, specificity 88%)[61]. It also has the ability to predict weaning outcome[38,61] and avoids the need for costly and technically-challenging invasive measurements[38,61]. I have previously conducted diaphragm ultrasound measurements in mechanically ventilated patients. Dr Hatem Soliman (consultant intensivist, Harefield Hospital), an expert in diaphragm ultrasound will provide comprehensive training.

Advice was sought from a health economist who advised cost-effectiveness of the intervention should not be measured in this study. However, the feasibility of collecting secondary outcomes measures such as the EQ-5D-3L would help inform cost-effectiveness in the definitive RCT.

Inclusion criteria

As per study 1.

Exclusion criteria

- Undrained pneumothorax/pneumomediastinum
- Pregnancy
- Considered unlikely to survive, or consultant assessment that the patient is not otherwise appropriate.

Consent procedures

As study 1

Sample size

The primary aim of the feasibility RCT is to estimate rate design parameters for the planning of a large-scale confirmatory RCT. With a sample size of 50, as recommended for feasibility studies with this aim[62] we will be able to estimate an overall drop-out rate to within a 95% confidence interval of at most +/-14%. This is based on a Wald confidence interval which is widest around a 50% estimate.

I recruited 5-6 patients a month and performed 1-7 three-hour studies per participant in my previous ICU IMT study at the Royal Brompton and Harefield hospitals[45] (42 beds). The participants' median duration on the study was 12 days before weaning of mechanical ventilation. The addition of the ICUs at Guy's and St Thomas' Hospitals and the Lane Fox Unit (68 beds) gives a larger pool of patients to recruit from in this study. With consideration of the training element of this award and different measurements conducted, I anticipate I can recruit and measure 4-5 participants per month over a 14-month period. My previous work shows I am in a strong position to achieve this.

Data analysis (Months 33-35)

All feasibility outcomes will be analysed using descriptive statistics (frequency, proportion and 95% CI). Between-group differences in clinical (secondary) outcomes will not be estimated. Instead, clinical outcomes will be analysed descriptively (and visually) to inform, in particular, the sample size calculation of a large-scale RCT. The proportion of missing clinical data as well as potential reasons for absence will be explored to inform a subsequent large-scale RCT.

Acceptability of delivering D-IMT

Acceptability of delivering treatment is an important component of the development of complex interventions. Semi-structured interviews, informed by a phenomenological approach will be undertaken with the costed physiotherapists (n=3-5) in the study to explore their experience of delivering D-IMT. Interviews will be audio-recorded, transcribed verbatim, anonymised and analysed using Braun and Clarke's six-step Reflexive Thematic Analysis phases[53].

Equity, diversity, and inclusion (EDI)

I am committed to research that actively supports and promotes EDI; every patient eligible to take part in the proposed research should have the same opportunity to participate. I have collaborated with the research EDI lead at GSTT in designing this proposal and will meet with them once per year, supplemented with email. The NIHR INCLUDE project guidelines and the Trust EDI impact assessment tool identified specific areas to improve participation opportunities, including: enabling surrogate approval for people who lack capacity to consent for themselves; overcoming language barriers by allocating funding for interpretation services for consent and research interventions; ensuring patient-facing materials and qualitative data collection tools are accessible by measuring readability scores, and by PPI group review. Relevant EDI data will be collected for all screened individuals and recorded on REDCap.

What do you intend to produce from your research?

Outputs for each study will be published open access in peer-reviewed journals to inform both physiotherapy and multidisciplinary colleagues of the findings. The most appropriate target journals will be discussed after analysis but due to the novel nature of this work and potential impact on practice, we would aim for journals such as the *American Journal of Respiratory and Critical Care Medicine* or *Critical Care*. Abstracts will be submitted to the European Conference on Weaning and Rehabilitation, European Respiratory Society Respiratory Failure and

Mechanical Ventilation Conference and national conferences. Abstracts will ensure the findings are shared widely and open opportunities for networking, discussion with world leaders, and future collaborations. Data will be managed within the Open Science Framework to maximise the integrity and value of the work.

How will you inform and engage service users and the wider population?

I will work closely with the Trust PPI lead and the project's PPI group to facilitate engagement with service users, services and the wider population. Results will be disseminated through the following:

- Participants and relatives from both studies will be invited to an in-person and video-conference feedback event to present and discuss findings.
- A lay summary developed with the project's PPI group will be shared in the ICU waiting areas and the hospital website.
- A dedicated social media account will be utilised to disseminate results to the research community and the wider population.
- The Brunel annual research festival, which engages with academics and the local community.
- I will engage with communication teams at GSTT, NIHR, charities (such as ICU Steps) and professional bodies, such as the Association for Chartered Physiotherapists in Respiratory Care (ACPRC) to identify further appropriate forums for dissemination.

How will your outputs enter our health and social care system or society as a whole?

The outputs of this research will be shared widely and freely with clinicians nationally and internationally as per the dissemination plan. This includes PPI dissemination, peer-reviewed journals and specialist physiotherapy groups. As an ACPRC editorial board member and a lecturer on the Advanced Clinical Practice MSc programme at Brunel, I have further opportunities to widely share the findings and change practice, such as enhancing clinicians' understanding of weaning and patient experience of IMT in ICU.

What further funding or support will be required if this research is successful?

If the feasibility RCT is successful, this programme of research has a clear path towards the design of a robust, fully powered definitive RCT comparing a diaphragm focussed, acceptable IMT regime to usual care which will measure its effects on clinical and patient related outcomes. This will be enabled by applying for funding through the postdoctoral NIHR ICA pathway or RfPB stream.

What are the possible barriers for further research, development, adoption, and implementation?

Despite a high proportion of physiotherapists using IMT in critical care, barriers to its use still exist. An international survey[37] and my own national survey (interim analysis) show 27 - 40% of physiotherapists report lack of evidence or clinical benefit prevents them using IMT in ICU. Evidence produced from this fellowship and the post-doctoral RCT aims to better inform the literature which, if positive outcomes are found, could improve adoption. Lack of equipment is commonly cited as a barrier to IMT use in ICU; a robust evidence base could help funding requests and subsequent IMT adoption.

What do you think the impact of your research will be and for whom?

In the short- and medium-term, study 1 will inform physiotherapists of a diaphragm-focussed and acceptable IMT load. Longer term, study 2 will inform a definitive trial to test the efficacy of this regime. If this regime is successful at improving outcomes such as weaning time, patients would benefit from a reduced duration of mechanical ventilation, ameliorating the negative effects of prolonged ventilation such as disability after discharge, activity limitations, impaired quality of life and cognitive impairment. Healthcare services would benefit from reduced ICU bed days and the associated reduced healthcare cost.

Project Management

I have an established supervisory team that I have worked alongside during my charity fellowship and PCAF, providing a strong history of effective supervision and project management. My supervisory team has significant experience in ICU research, mixed methods methodology, physiology, point of care ultrasound and previous PhD supervision. I will meet with my academic supervisors monthly in addition to individual meetings and email correspondence. I will meet quarterly with all supervisors together. Steering group meetings will be held 1-2 times per year with me, principal lay advisor and my supervisory team. As the project lead, I will take responsibility for day-to-day management of the project including: recruitment and consent; data collection and analysis with support of a statistician; regular updates with key stakeholders (Research and development department, REC, NIHR, PPI group). GCP guidance will be adhered to throughout the project. I have experience in financial management of charity and NIHR grants and will be supported by the GSTT research finance department during this award.

Ethics

Appropriate ethical approvals will be sought via the Health Research Authority, Research Ethics Committee, and the host organisation. Both studies will be registered on clinicaltrials.gov. The oesophageal catheter used in study 1 is used in clinical and research settings, is CE and UKCA marked and replaces the participant's NG tube. IMT has been shown to be safe in mechanically ventilated patients[14,23,46]. Application for ethical approval of study 1 will be made as soon as the fellowship decision is communicated, with the aim of commencing the study in the first month of the fellowship.

Success criteria

Development of a diaphragm-focussed, patient-acceptable IMT regime; completion of all study stages within budget and timeframe; dissemination of the findings as above; completion of intended training programme; completion and award of a PhD.

Risks

Risk 1: Delay in obtaining ethics approval.

Mitigation 1:

- Study 1 approval will be sought prior to commencement of the fellowship.
- A four-month period for ethical approval of study 2; Appropriate areas of the protocol and IRAS form will be drafted before this time.
- I have an established working relationship with the trust research and development office who are aware of my proposed fellowship deadlines.

Risk 2: Failure to recruit; unforeseen circumstances mean the studies may not be delivered as expected. Failure to recruit in study 1 could delay or prevent study 2 commencing.

Mitigation 2:

- Recruitment will be closely monitored and actively managed in both studies.
- Recruitment rates have been calculated from similar research I have conducted at RBHT[45], the addition of GSTT sites increases the pool of patients to recruit from. The Royal Free Hospital has agreed to be an additional site if required.
- Extensive PPI input has shown this research is acceptable to patients, minimising the risk of widespread non-enrolment.

Risk 3: Inconclusive or little qualitative data from study 1

Mitigation 3: Rigorous methodology; Extensive PPI and speech and language therapist input, detailed discussions with my supervisory team and up to three research occasions per participant mean we have

maximised my ability to gain rich qualitative data.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I worked closely with the Trust PPI lead to form a specific PPI group for my research interest and this project. The PPI group consists of one relative of an ICU survivor, and six ICU survivors, all of whom experienced prolonged mechanical ventilation and experienced physical rehabilitation in ICU. Five members performed Inspiratory Muscle Training (IMT) when mechanically ventilated. Three of these members participated in my previous study, and therefore have specific experience of performing IMT at four different loads in one research occasion. The complex medical and psychological burden of being critically unwell means this group has been invaluable in forming this proposal. PPI was fully funded in my PCAF, and I have met with the group five times in preparation of this application.

Influence and changes to proposal

The group understood the importance of the proposed research and felt they or their next of kin would consent for the research. They advised to increase the rest period between IMT loads, as they felt participants would experience fatigue and breathlessness after IMT. The rest would enable them to recover fully, and to participate in the qualitative data collection.

The group gave specific guidance on the scale used to measure breathing difficultly and unpleasantness by suggesting simple anchor points using a number rating scale to help cognitively impaired or fatigued patients understanding of the scale.

The group advised that although placement of the oesophageal catheter can be uncomfortable, they realised the importance of it for the research, and that a single tube used for both feeding and research would be much more favourable; the protocol was changed accordingly. The group gave detailed contributions to the plain English summary.

The group advised heavily on the qualitative data collection and offered first-hand accounts of their changing cognition throughout their ICU stay. Due to this, they suggested 1-3 research occasions per participant in study 1 to improve the richness of the qualitative data and account for variable engagement. The group gave key words describing their experience of different IMT loads and breathlessness, words which will be used in the qualitative data collection 'toolbox'.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Current PPI group members have found the experience rewarding and productive; five out of the seven members have agreed to continue the role throughout this fellowship. An additional two members will be recruited in the coming months.

One member (LK), a previous research participant with first-hand experience of different IMT loads when mechanically ventilated, has agreed to be PPI lead and will attend steering group meetings to ensure a direct link between the PPI group and the research team. LK and I will and attend a national ICU conference where he will be given the opportunity to co-present findings.

Training and support

I will work closely with the Trust PPI lead to ensure I continue to follow the UK standards for public involvement in research. Briefly, the group works well together and all contributions are valued and respected; individual support and guidance will be provided by the Trust PPI lead; there is allocated funding for their involvement and to attend PPI training; I will meet separately with the PPI lead prior to meetings to discuss findings to support him in his role.

Members of the PPI group have been given a role description. The entire group agreed that meetings will take place using video conferencing to ensure equal access (all members are computer literate but live in different geographical locations with varied levels of physical function).

Continued involvement in proposed research

The group will meet three times per year, with email updates between meetings. Specific activities will include:

- Discuss and clarify the qualitative questioning to be used in study 1 (funded in PCAF).
- Advise on study 1 data interpretation.
- Review lay summaries for ethics applications.
- Review all patient facing documents.
- Help produce dissemination material and provide guidance on dissemination methods.

Two members of the initial PPI group are unable to continue participation due to family commitments or medical concerns, which has resulted in a predominantly white group. I recognise the absence of underserved groups and am actively working with the Trust PPI lead to increase diversity in our aim to recruit more members, in line with NIHR's Under-served Communities Programme.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
n/a

8. Training & Development and Research Support

Proposed training and development programme

This training plan has been developed using the VITAE researcher development framework with my supervisory team to build on my existing skills. The proposed project has clear potential for benefiting patients, while enabling broad training and development for my progression as a clinical academic leader. This training plan will equip me with the skills and knowledge to conduct the proposed research and subsequent post-doctoral research.

Proposed formal study

As a doctoral researcher at Brunel, I will complete the required progression reviews and training. The formal training element will primarily be provided by the Brunel Doctoral School and cover topics from the four domains of the researcher development framework. The following courses will further my development:

- **Year one:** Doctoral research: getting started; embedding EDI into research; academic resilience; critical thinking and writing.
- **Year two:** ANOVAs in SPSS; delivering conference presentations; doctoral research: moving forward.
- **Year three:** Doctoral research: completing successfully; preparing thesis submission.

As an employee of Guys and St Thomas' NHS Foundation Trust I have access to support, events and masterclasses run by the Imperial and King's Clinical Academic Training Office. Their annual research symposium will give me opportunities to present my work and network.

Additional academic training and development to support my development as a researcher and academic leader

This plan links to specific methodology, statistics, data analysis and interpretation skills required at key milestones of my project.

Year one *(study 1):*

- **Introduction to qualitative interviews** (University of Oxford, 1 day). Practical and theoretical elements of qualitative research interviews – refreshes and builds on skills learnt on the Brunel qualitative research methods MSc module.
- **Qualitative data analysis: Theory and Practice** (National Centre for Research Methods, 1 day): To develop my skills in qualitative data preparation, coding and identifying themes.
- **Mixed Methods Design and Analysis** (National Centre for Research Methods, 2 days): Design, sampling, analysis and integration of qualitative and quantitative methods.
- **Facilitation: how to get the most out for your PPI activities** (UCL, 2 hours). Will improve my skills in identifying research priorities and designing protocols with PPI members.
- **Increasing Participation of Ethnic Minorites in Health and Social Care Research** (Centre for Ethnic Health Research / NIHR, eLearning): I recognise the need to increase my awareness of EDI in research; this training will help me achieve more diverse ethnic representation in my research.
- **Inside a Clinical Trials Unit** (Queen Mary University London Pragmatic Clinical Trials Unit, 9 months, 4 hours/month) This extensive course provides learning and mentorship in a live setting. I will gain experience in the design, conduct and governance of pragmatic clinical trials by observation and one-on-one tutor supervision (12+ hours). This will equip me with the skills to conduct the feasibility RCT in study

2, and further post-doctoral research.

Year 2 *(study 1 and 2):*

- **Mixed methods analysis and integration for health research** (Warwick Medical School, 4 days): Principles of mixed methods analysis: transformation, synthesis, integration, and analysis.
- **Supervised diaphragm ultrasound scans** (three months, 4 hours/week): Supervised by Dr Soliman (Consultant Intensivist, Harefield Hospital). As an internationally renowned expert in ICU point of care ultrasound, Dr Soliman will develop my skills in diaphragm ultrasound in preparation for study 2.
- **Randomised controlled trials course: A guide to design, conduct, analysis, interpretation and reporting** (NIHR Oxford Biomedical Research Centre, 5 days): This course will provide me with a thorough grounding in RCT methodology and the skills to conduct a successful RCT.

Year 3 *(study 2):*

- **Writing and publishing qualitative research** (UCL, 1 day): This course will support me in describing and writing up qualitative data.
- **Introduction to linear and logistic regression models** (University of Bristol, 5 days). I have some experience in regression analysis; this course will help with more complex data analysis from study 2 and beyond.
- **Leadership development** (NIHR eLearning, 3 hours)

Additional training and development to support my personal and professional development as a practitioner and practice leader

This training will facilitate my development as a clinical academic and leader in critical care:

- **Advanced course in acute respiratory failure and mechanical ventilation** (European Society of Intensive Care Medicine, 2 days) Directly linked to my aspiration to be a leader in ventilator weaning. Covers advanced assessment and support of ventilated patients.
- **Emerging clinical leaders** (King's fund, 6 days). This course will help my understanding and subsequent development of my leadership style as a clinical academic. These skills will help improve team collaboration, wellbeing, and importantly, patient care.

I will attend multidisciplinary conferences to explore up-to-date research and network with clinical academic leaders.

- **Intensive Care Society State of the Art Conference** (2026): Clinical learning, dissemination of findings with PPI lead at national conference.
- **10th European Conference on Weaning and Rehabilitation** (2026): Clinical learning, networking, dissemination of findings.
- **Association of Chartered Physiotherapists in Respiratory Care** (2027): Clinical learning, networking.
- **European Respiratory Society Respiratory Failure and Mechanical Ventilation Conference** (2027): Clinical learning, networking, dissemination of findings.

Details of practice to be undertaken during the fellowship

My professional practice and development plan is aimed at maintaining and developing my skills to become a physiotherapy leader and expert in rehabilitation, mechanical ventilation, and weaning in in critical care. I will complete a four-month secondment (20%) at the Lane Fox Unit, a world leading centre in ventilator weaning and rehabilitation, where I will advance my clinical practice with multidisciplinary leaders in the field.

To progress my role as a leading clinical academic I will spend three individual weeks (full-time) at Coventry University Hospital under the supervision of Professor McWilliams to advance my clinical and academic skillset.

I will visit Professor Daniel Langer in Leuven, Belgium for a research and clinical visit. Professor Langer's lab conducts research in rehabilitation and inspiratory muscle training in critical care. He is an international expert on IMT in critical care, with >100 peer reviewed articles.

Throughout the fellowship I will work clinically at the Brompton and Harefield Hospitals to maintain and develop my specialist skills in critical care physiotherapy. I will continue supporting colleagues with research and service development to build research capacity within the clinical team.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Dr Vicky MacBean

Institution	Brunel University London

Position	Senior Lecturer

Description of support provided
Dr MacBean is an academic physiotherapist with an international profile in the field of respiratory muscle function and physiology and perception of respiratory effort. She worked clinically in the UK and New Zealand, specialising in paediatric respiratory and critical care physiotherapy, before moving into academia. She undertook doctoral and postdoctoral research at King's College London, working alongside world leaders in the field (Professors Polkey, John Moxham, Anne Greenough, Nicholas Hart, Dr Gerrard Rafferty). She moved to Brunel University in 2018 as Lecturer in Physiotherapy, becoming Senior Lecturer in 2023. Dr MacBean has been instrumental in progressing my clinical academic research career since the planning and award of my charity fellowship in 2020. I have met with her on a regular basis since, depending on the needs of the project at the time, and she is the primary supervisor for my PCAF. It is acknowledged that Dr MacBean's formal doctoral supervisory experience is still growing. She nevertheless has extensive experience of formal and informal research supervision (MSc, MRes and MD/PhD), both as a doctoral/postdoctoral fellow and as a tenured academic. She is currently or has been a member of the supervisory team for five doctoral researchers at Brunel, and on progression panels for a further two doctoral researchers. She has longstanding working relationships with Professors Kilbride (since 2019) and Polkey (since 2009), and the track record of successful joint project completions and publications attests to the quality of the collaboration and the supervisory team's support for Dr MacBean as primary supervisor.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Professor Cherry Kilbride

Institution	Brunel University London

Position	Professor of Neurorehabilitation and Applied Clinical Research

Description of support provided
Professor Cherry Kilbride is a physiotherapist and Professor of Neurorehabilitation and Applied Clinical Research at Brunel University London, and the Lead Allied Health Professional for Research at the Royal Free London NHS Foundation Trust. Professor Kilbride also has a strategic commitment for the development of AHP academic clinicians at the Trust. Professor Kilbride joined Brunel in 2007 with twenty years' experience of being a respected healthcare leader, manager and clinician. Professor Kilbride is an expert in mixed methods research with over 70 peer reviewed publications and has a sustained record of attracting and delivering grants to time and target in excess of £1.8m. Professor Kilbride has seen seven PhD students through to completion, has six ongoing doctoral students (including NIHR) and supervised over 50 MSc dissertations. Professor Kilbride's mentorship experience spans NIHR Advanced, Doctoral, Predoctoral Fellows to Interns. She has also overseen 7 therapists on the NIHR Associate Principal Investigator role. Professor Kilbride will provide close mentorship and support to Dr MacBean; they are co-located at Brunel. I first met Professor Kilbride in the early stages of my charity fellowship, and she has since helped my progression as a clinical academic; her capacity as a mentor in my PCAF quickly turned to academic supervisor for my DCAF due to her expertise in mixed methods research. Since taking this role Professor Kilbride, Dr MacBean, Professor Polkey and myself have met regularly, proving her capacity to supervise me, and for the supervisory team to collaborate effectively.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Professor Michael Polkey

Institution	The Royal Brompton and Harefield Hospitals

Position	Consultant physician and honorary professor (Imperial College London)

Description of support provided
Professor Polkey was appointed as a consultant in respiratory medicine at the Royal Brompton Hospital in 2000. He is now interim director of research at the Royal Brompton and Harefield Hospitals in addition to being chair of the care group dealing with home non-invasive ventilation and weaning. Professor Polkey's expertise aligns well with my DCAF program of work; he is an expert in respiratory physiology, beginning with his own thesis in 1997. He now heads the respiratory muscle laboratory at the Royal Brompton Hospital. His clinical practice involves the assessment of patients with advanced COPD for rehabilitation, complex weaning in critical care, and home ventilation. Professor Polkey has vast research experience with >400 publications. He is or has been associate editor of the *European Respiratory Journal*, *Thorax* and *Clinical Science*. He has supervised or co-supervised 34 students to PhD (or occasionally MD), of whom five are now Professors of Medicine. Professor Polkey's extensive experience of doctoral supervision and internationally leading research places him in a strong position to support both the programme of research and the doctoral supervision element. Professor Polkey fully supports Dr MacBean as my primary supervisor and has offered mentorship and support to her as required. Professor Polkey has been a core member of my supervisory team since 2020, with a proven capacity to supervise me throughout this time; we have met every 2-4 weeks for supervision and project meetings with my supervisory team, and he is extremely responsive to emails, normally responding within hours.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Professor David McWilliams

Institution	University Hospitals Coventry and Warwickshire NHS Trust and Coventry University

Position	Professor of Critical Care and Rehabilitation

Description of support provided
Professor McWilliams is a Professor of Critical Care and Rehabilitation within University Hospitals Coventry and Warwickshire NHS Trust and Coventry University. He has extensive clinical experience in critical care physiotherapy and complex rehabilitation. He is Chair of the Intensive Care Society National Rehabilitation Collaborative and has contributed to the development of national guidance for critical illness rehabilitation (NICE CG83, NICE QS158). David's research interests focus on rehabilitation and recovery following a period of critical illness, with >50 peer reviewed publications and over £5 million in grant income as lead applicant or within research teams. David is a recognised international expert in critical care physiotherapy and rehabilitation, regularly presenting nationally and internationally. David actively supports and mentors clinical academic AHPs from across the UK, including 2 current NIHR DCAF students and 2 NIHR PCAF awards. David will provide practice supervision specifically around the development and evaluation of complex rehabilitation interventions within an NHS context. As an established Senior Clinical Academic and Clinical Professor, he will also provide direction and mentorship on the development of a senior clinical academic role. With an established supervisory relationship that started over one year ago, David has a thorough understanding of my professional training and development needs, where we have met individually or with my academic supervisors every 2-3 months. Alongside our research collaboration, he will also use his experience to provide direct support and mentorship in my development towards a clinical academic role. We will meet online bi-monthly to discuss my progress.

PRACTICE SUPERVISOR 2

Name	Dr Hatem Soliman

Institution	The Royal Brompton and Harefield Hospitals

Position	Consultant Intensivist in Cardiothoracic Intensive Care

Description of support provided
Dr Soliman is a Consultant in Cardiothoracic Intensive Care at Royal Brompton and Harefield Hospitals, visiting Professor at Mayo Clinic, Florida and Honorary Senior Lecturer at the School of Cardiovascular Sciences and Medicine at King's College London. Dr Soliman has >25 peer reviewed publications in point of care ultrasound in ICU and is Associate Editor of the Critical Care Ultrasound Journal and Frontiers in Cardiovascular Imaging. He is lead editor of a textbook entitled 'Cardiopulmonary Point of Care Ultrasound', published in 2023. Dr Soliman has an international profile in teaching critical care ultrasound and was awarded the prestigious excellence in education awards from the European association of cardiovascular imaging in 2022.

Dr Soliman has trained >200 medics, six physiotherapists and nine nurses in Focused Ultrasound in Intensive Care (FUSIC) internationally. He developed a multidisciplinary point of care ultrasound training course where diaphragm ultrasound is a core part of the programme. The programme was shortlisted for the best new educational programme in the Intensive Care Society awards 2024.

Dr Soliman will provide extensive teaching and supervision to refine my skills in diaphragm ultrasound before study 2 commences. This will include four hours per week (over three months) of combined and independent diaphragm scanning on mechanically ventilated patients. Dr Soliman and I have an established relationship (>4 years), where he has previously supervised me in conducting lung and diaphragm ultrasound. Dr Soliman will provide support throughout the fellowship with bi-monthly supervisions and additional support as required during study 2.

Collaborations

Royal Brompton and Harefield Hospitals PPI team

In addition to extensive ongoing and planned PPI involvement in the proposed project, I will continue to collaborate with the PPI lead at the Trust, Zahra Aden, for ongoing advice and support. With this, I will present findings to the established respiratory PPI group at the Royal Brompton and Harefield Hospitals (RBHT) and gain opinions and advice on any further dissemination strategies they might suggest.

Post-graduate education and supervision, Brunel University

I understand a career as a clinical academic leader means contribution towards the training and development of both future clinical academics and clinicians. My experience teaching on MSc modules, supporting two MSc students, and supporting clinicians with research has begun this process. An important element of my development as a clinical academic leader will be for me to support further MSc projects; I will collaborate with my academic supervisory team and other research leaders at Brunel university to facilitate this. The number of projects I support will be discussed with my supervisory team to ensure it does not impact on timescales within this proposed application. I will also take further opportunities to teach on appropriate MSc advanced clinical practice modules.

Lee Bolton, clinical lead, speech and language therapist, RBHT ICU

I will continue my collaboration with Lee Bolton who will offer support and advice before and during study 1. Lee Bolton has extensive expertise in the assessment and management of communication critical care. This includes patients with tracheostomy and ventilator-dependency who have concomitant laryngeal and upper airway complications secondary to critical care interventions. He also lectures on critical care postgraduate modules for Brunel University and University of West London.

In collaboration with the projects PPI group, his specialist input will support development of the written communication boards for study 1 of the proposed research. He will provide expert advice and treatment for complex patients recruited to study 1, to ensure participants are able to communicate as effectively as possible for the qualitative data collection proposed.

International Collaborations

Professor Stephen Rees and Associate Professor Dan Karbing

I will continue my collaboration with Professor Rees and Associate Professor Karbing who head the Respiratory

and Critical Care group at Aalborg University, Denmark. They are experts in the physiological response to exercise and ventilatory weaning in mechanically ventilated patients. They have published lab data on diaphragmatic activation measured by electromyography (EMG) during exercise, and respiratory muscle fatigue during loaded breathing. They are co-authors on my published work and one manuscript under review. I am visiting them in November 2024 for a research visit as part of my PCAF to analyse lab data collected during my charity fellowship. Professor Karbing has provided advice on the healthy EMG IMT study at Brunel (results of which inform this DCAF) and on elements of the proposed work in this fellowship. Our continued collaboration during my DCAF via video conferencing and email will ensure journal outputs for our healthy IMT data and potential post-doctoral research collaborations.

Associate Professor Ulrike Pielmeier

I will continue a growing collaboration with Associate Professor Ulrike Pielmeier in the Department of Health Science and Technology at Aalborg University, where I was initially approached to advise on an early mobilisation system in critical care. We now plan to conduct a short, one month study at Harefield Hospital in January 2025, measuring agreement of an early mobilisation decision system versus clinician decision to mobilise in critical care. This work will be concluded before the proposed start date of my DCAF. Results from this study may inform subsequent studies in my post-doctoral career, and further strengthens links with Aalborg University.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

Tim has been at Royal Brompton and Harefield hospitals (RBHH) for six years and is already recognised as a potential future leader in research. During this time, he has completed a ten-month competitively awarded Royal Brompton and Harefield Hospitals Charity (RBHC) Fellowship in Inspiratory Muscle Training (IMT), which is Tim's area of interest for this DCAF. Tim recruited his planned sample size in record time and was able to then double his recruitment target. He used the RBHC Fellowship to establish links with key collaborators in the UK and abroad who will be integral to his research in the future. Tim was also personally selected to take part in an Imperial College London 'Starting Out in Research' course, which he now mentors on. He is currently completing an NIHR PCAF which is the foundation for this DCAF.

Royal Brompton and Harefield Hospitals (RBHH) have a long-standing successful partnership with Brunel University London (Brunel), having joined the Brunel Partners Academic Centre for Health Sciences (BPACHS), and we believe we offer the ideal environment for Tim to advance as a practitioner academic. As a partnership, we also support a lecturer (Lewis) on an ongoing NIHR Research for Patient Benefit grant. Our links are further strengthened through Dr Claire Nolan who is completing an NIHR AF across our organisations.

Clinically, Tim will be hosted by the Rehabilitation and Therapies (R&T) Directorate of RBHH. Tim's application comes with strong support from the Directorate and the Trust as it aligns with our aim to develop NMAHPP-led research. The R&T Directorate currently has a significant number of ongoing fellowship recipients and we have been able to facilitate these through successful backfill. We value our fellows and Tim has full support from the critical care therapy management team, and the R&T team, to maintain clinical practice and progress as one of our future practitioner academic leaders.

RBHH have an active and excelling allied-health professional (AHP) community with regular AHP-focused for a, giving our researchers an opportunity to come together to discuss research and offer peer-support and fellowship guidance to one another. As a physiotherapist, Tim will be well supported by an expanding team of independent researchers in the same profession who have completed PCAFs (x2) and/or are undertaking or have recently completed NIHR DCAFs/CDRFs (x6), and those who have secured and are leading on their own research grants

as Principal Investigators (x4). Two of Tim's peer group (Grillo and Balls) have successfully moved within the NIHR ICA pathway from PCAF to DCAF, as Tim hopes to, and will be an invaluable support to him on his practitioner academic journey.

Academically, Tim will be hosted within the Department of Health Sciences at Brunel. The department has a strong focus on interdisciplinary research and inter-professional learning, hosting multi-professional disciplines (Physiotherapy, Occupational therapy, Nursing, Social Work, Public Health and Health Economics). Tim will have access to eight different Research Centres and will specifically be attached to the Centre for Physical Activity in Health and Disease. The department is supported by a dedicated Director of Postgraduate Research, Dr Wendy Martin, who promotes excellence in doctoral education, training and engagement. Tim will benefit from the departments' prioritisation of a values-based approach to doctoral research which ensures a positive and ethical organisational culture that facilitates academic citizenship. Tim will be further supported by the Graduate School through various training programmes and will have a dedicated advisor who will monitor and support his career development.

Brunel's Physiotherapy Division has a large number of research-active academics working across a range of clinical and laboratory-based topics and has a wealth of qualitative, quantitative and mixed-methods experience. We hold a monthly divisional research meeting, at which doctoral researchers are actively encouraged to participate, presenting study plans, progress, and research findings in a friendly and supportive environment. These meetings are also an opportunity to gain exposure to broader research topics, supporting Tim's development as a well-rounded practitioner academic. Tim has already been a valuable contributor as a guest lecturer on our Advanced Clinical Practice MSc programme; he will have the opportunity to further develop his teaching practice during his doctoral studies. This may include assisting in supervision of MSc research projects.

The Department has a commitment to research excellence and was highly evaluated on the Research Excellence Framework (REF) 2021. In the Unit of Assessment 3 (Allied Health Professions, Dentistry, Nursing and Pharmacy) 84% outputs were rated 3* OR 4*, with 42% as 4* ("World leading"), 75% of impact case studies were rated at 3* or 4*and 100% of the research environment was rated as 3 or 4*. Tim will therefore be hosted and supported by a department that is research-intensive and progressive in designing, conducting and translating research into clinical practice and wider policies.

As one of the largest employers in the NHS, GSTT (of which RBHH is part) has an incredibly diverse workforce and takes proactive measures at all levels to make equality, diversity, and inclusion (EDI) a part of everyday business. A Positive Action Charter has been established for managers and senior leaders to pledge their commitment to helping GSTT embed a culture of EDI across the organisation. Other EDI initiatives, summarised on an EDI toolkit, include inclusion agents, rainbow badges, sunflower lanyards, staff health passports, workshops, Positive Pathways development programme, mandatory EDI training and many more staff networks.

Brunel is a research intensive and civic university, renowned for its commitment to diversity, to widening participation and with a social mission to make a positive and meaningful difference to people's lives and the environment, locally and globally (see https://www.staff.brunel.ac.uk/documents/Wielding-Brunels-World-class-Difference.pdf). Tim's host department achieved an Athena Silver Award in 2023, demonstrating commitment to gender equality in education and research. Brunel research processes are guided by the EDI principles, supported by a dedicated Associate Dean for EDI, Dr Mike Thomas, a social work researcher.

With our existing and developing links, RBHH and Brunel are committed to ensuring that Tim receives the required time, support, training and networking opportunities to allow him to successfully complete his DCAF objectives.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Timothy Jenkins

Employing organisation	Guy's and St Thomas' NHS Foundation Trust

Grade	AfC

Starting point on scale	44621.21

Increment Date	01/04/2025

Justification of Costs
Lead applicant, 1WTE. Salary has been calculated and verified by Research Finance at Royal Brompton Hospital

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	44621.21
Geographical Weighting (£)	7901.69
Other allowances (£)	
Superannuation (£)	10861.74
Nat. Insurance (£)	7904.7
Current Annual Costs (£)	**71289.34**

2

Staff type	Shared Staff

Name	TBC

Employing organisation	Guy's and St Thomas' NHS Foundation Trust

Role on this research	Physiotherapist time to deliver IMT intervention in feasibility RCT (study 2)

Grade	AfC B6

Starting point on scale	38386.49

Increment Date	01/04/2026

Justification of Costs
This staff time is required to ensure allocation concealment and intervention binding in the feasibility randomised controlled trial in study 2. To account for leave and sickness, two Physiotherapists in each ICU (total 4-8) will randomise participants and deliver the D-IMT intervention which takes 15 minutes per session. Median time in study: 12 days, 50 participants. Approximately 240 hours staff time

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	38386.49
Geographical Weighting (£)	7677.3
Other allowances (£)	0
Superannuation (£)	9525.99
Nat. Insurance (£)	6932.6
Current Annual Costs (£)	**62522.38**

3

Staff type	Shared Staff

Name	Thomas Hamborg

Employing organisation	Queen Mary University of London

Role on this research	Statistician

Grade	6RES / 042

Starting point on scale	57015

Increment Date	01/04/2025

Justification of Costs
Statistician

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	57015

	Total (£)
Geographical Weighting (£)	0
Other allowances (£)	0
Superannuation (£)	8710.4
Nat. Insurance (£)	7620.31
Current Annual Costs (£)	**73345.71**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mr Timothy Jenkins
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Lead applicant / Fellow's salary.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	44621	45067	45518	135206
National Insurance	7905	7984	8064	23953
Superannuation	10862	10970	11081	32913
Geographical weighting / Other allowances (if applicable)	7902	7981	8061	23944

Staff Type: Other
Other Type: Shared Staff
Name: **Physiotherapist to deliver IMT intervention in study 2**
Type of cost: NHS
% Full-time on this research: 10.0
Total months on this research: 14.0
Justification of Cost: This staff time is required to ensure allocation concealment and intervention binding in the feasibility randomised controlled trial in study 2. To account for leave and sickness, I will train two Physiotherapists in each ICU (total 4-8) to randomise and deliver the D-IMT intervention which takes 15 minutes per session. Median time in study: 12 days, 50 participants. Approximately 240 hours staff time

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	0	3647	3684	7331

Staff Type: Other
Other Type: Shared Staff
Name: **Thomas Hamborg**
Type of cost: HEI
% Full-time on this research: 5.0
Total months on this research: 36.0
Justification of Cost: Statistical support

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	3749	3855	3964	11568

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	9254	9254
Total NHS	216016	0	7331	223347
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: 10th European Conference on Weaning and Rehabilitation
Justification of Cost: This conference is a specialist European conference on weaning and rehabilitation in Intensive Care. It attracts world leading delegates and speakers. It will provide opportunities for clinical learning, networking and dissemination of findings.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	160	0	160

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: 10th European Conference on Weaning and Rehabilitation
Justification of Cost: 3 nights accommodation at European city (location for conference not finalised)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	400	0	400

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: 10th European Conference on Weaning and Rehabilitation
Justification of Cost: Approx cost of flight

	Year 1	Year 2	Year 3	Total Cost (£)
	0	320	0	320

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: Intensive Care Society State of the Art Conference (UK)
Justification of Cost: I will attend the Intensive Care Society State of the Art Conference (UK) - a multidisciplinary conference for ICU professionals. I will have the opportunities for clinical learning and networking. I plan to disseminate findings to date with the PPI lead, if the PPI lead agrees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	360	0	360

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: Intensive Care Society State of the Art Conference (UK)
Justification of Cost: 2 nights accommodation in UK city (location of conference not confirmed)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	180	0	180

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Intensive Care Society State of the Art Conference (UK)
Justification of Cost: Average cost of return train journey to UK city

	Year 1	Year 2	Year 3	Total Cost (£)
	0	140	0	140

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: Association of Chartered Physiotherapists in Respiratory Care
Justification of Cost: I will attend the Association of Chartered Physiotherapists in Respiratory Care conference in Year 3 which will provide opportunities for clinical learning and networking with physiotherapists specialising in ICU and respiratory.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	100	100

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: Association of Chartered Physiotherapists in Respiratory Care
Justification of Cost: 2 nights accommodation in UK city (location of conference not confirmed)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	180	180

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Association of Chartered Physiotherapists in Respiratory Care
Justification of Cost: Cost of return train journey

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	140	140

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: European Respiratory Society Respiratory Failure and Mechanical Ventilation conference
Justification of Cost: I will attend the European Respiratory Society Respiratory Failure and Mechanical Ventilation. This conference in my specific area of interest will provide me with opportunities for clinical learning, networking and dissemination of findings to date. The conference is free to attend for Physiotherapists

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: European Respiratory Society Respiratory Failure and Mechanical Ventilation conference
Justification of Cost: 4 nights accommodation in European city

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	400	400

Type of cost: NHS
Description: Journey Costs
Conference: True

Item Description: European Respiratory Society Respiratory Failure and Mechanical Ventilation conference
Justification of Cost: Approx. cost of flights

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	200	200

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	0	1560	1020	2580

Equipment

Type of cost: NHS
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 100
Expected life time: 60
Justification of Cost: I will need a laptop for data collection as I will be moving between ICU locations and working remotely on occasion.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	1000	0	0	1000

Type of cost: NHS
VAT relief to be applied: False
Description: Tablet
Price excluding VAT £: 300
Expected life time: 60
Justification of Cost: I require a basic android tablet to run the inspiratory muscle device (the PowerBreathe KH2 inspiratory muscle trainer)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	300	0	0	300

Type of cost: NHS
VAT relief to be applied: False
Description: Dictaphone
Price excluding VAT £: 150
Expected life time: 60
Justification of Cost: I will require a dictaphone in order to record qualitative interviews.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	150	0	0	150

Type of cost: NHS
VAT relief to be applied: False
Description: NAVA module for diaphragm EMG measurments
Price excluding VAT £: 4950
Expected life time: 36
Justification of Cost: Required for measurement of inspiratory muscle activation of the diaphragm using the esophageal multipair EMG catheter (NAVA, Getinge, Sweden). I have one module I am using for my PCAF study. Two modules will enable easier data collection for study 1 between central and outer London ICU locations within Guy's and St Thomas' NHS Foundation Trust, as the modules are attached to a bulky ventilator

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	4950	0	0	4950

Type of cost: NHS
VAT relief to be applied: False
Description: Powerbreathe KH2 inspiratory muscle trainer
Price excluding VAT £: 1499
Expected life time: 72
Justification of Cost: Required for study 1 and 2 to administer inspiratory muscle training (IMT)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	1499	0	0	1499

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	7899	0	0	7899

Consumables

Description: Oesophageal multipair EMG cathaters (NAVA, Getinge, Sweden)
Type of cost: NHS
Please describe fully: Oesophageal multipair EMG cathaters (NAVA, Getinge, Sweden)
Justification of Cost: NAVA multipair EMG catheters for measurement of diaphragm activity in study 1 (n=40). Costed for 50 to allow for accidental removal by patients or the clinical team within the study period for participants.

	Year 1	Year 2	Year 3	Total Cost (£)
	9480	0	0	9480

Description: Kendall Arbo disposable EMG electrodes
Type of cost: NHS
Please describe fully: Kendall Arbo disposable EMG electrodes
Justification of Cost: 1000 EMG electrodes for extra-diaphragmatic EMG measurements. n=50, up to 120 study occasions, x8 for each occasion (disposable)

	Year 1	Year 2	Year 3	Total Cost (£)
	268	0	0	268

Description: Ultrasound gel
Type of cost: NHS
Please describe fully: Ultrasound gel
Justification of Cost: Ultrasound gel x6 250ml for diaphragm ultrasound training and use in study 2

	Year 1	Year 2	Year 3	Total Cost (£)
	0	15	15	30

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	9748	15	15	9778

Patient and Public Involvement

Description: Payment & Reward
Type of cost: NHS
Please describe fully: Payment & Reward
Justification of Cost: Fees to lay advisors (attending lay advisory group meetings), 7 lay advisors, 3 meetings per year, 1.5 hour meetings, 3 years = £2,362.5 Fees to lay advisors (work between lay advisory group meetings), 7 lay advisors, 3 items of work per year, 1.5 hour per activity, 3 years, = £2,362.5 Fees to lay advisors (attending steering group meetings), 1 lay advisor, 2 meetings per year, 2 hour meetings, over 3 years = £300 Fees to lay advisors (work between steering group meetings), 1 lay advisor, 2 items of work per year, 1.5 hour per activity, over 3 years, = £225 Section Total: £5,250.00

	Year 1	Year 2	Year 3	Total Cost (£)

	1750	1750	1750	5250

Description: Training and courses
Type of cost: NHS
Please describe fully: Training and courses
Justification of Cost: ICL PPI training (time for lay advisors to undertake), 7 lay advisors, 4 sessions, 2 hours per session = £1400 UCL PPI facilitation training course, 1 lay advisor, 1 session = £40 UCL PPI facilitation training (time taken for PPI lead to undertake), 1 lay advisor, 1 session, 2.5 hours = £62.50 Section Total: £1,502.50

	Year 1	Year 2	Year 3	Total Cost (£)
	501	501	501	1503

Description: Events/dissemination
Type of cost: NHS
Please describe fully: Events/dissemination
Justification of Cost: Patient dissemination event, 1 event, costs associated with catering, printing/other admin = £150 PPI lead attending conference (Intensive Care Society's, State of the Art Congress), 1 lay advisor travel and registration fee = £600 Section Total: £750.00

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	750	750

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	2251	2251	3001	7503

Other Direct Costs

Description: Poster printing
Type of cost: NHS
Please describe fully: Poster printing
Justification of Cost: Added for the years I am attending conferences

	Year 1	Year 2	Year 3	Total Cost (£)
	0	70	70	140

Description: Transcription of qualitative interviews
Type of cost: NHS
Please describe fully: Transcription of qualitative interviews
Justification of Cost: For study 1 - £1.50 per minute, 30 minutes max per participant (x120). For study 2 - physiotherapist (implementing D-IMT) qualitative interviews, £1.50 per minute, 60 minutes max per interview for up to five interviews. Accurate and efficient transcription of patient interviews in study 2 is essential. I am not experienced in transcription - the most appropriate and efficient way of doing this is via a transcription service. An average cost of £1.50 per minute of transcription for the maximum of 120 qualitative data collection interviews lasting 30 minutes, and 5 staff interviews lasting 60 minutes has been costed here.

	Year 1	Year 2	Year 3	Total Cost (£)
	5400	0	450	5850

Description: Database
Type of cost: NHS
Please describe fully: Database
Justification of Cost: RedCAP database for data collection and to support analysis. Annual licence fee requested

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	1500	1500	4500

Description: Translator / interpreter time
Type of cost: NHS
Please describe fully: Translator / interpreter time

Justification of Cost: 9 hours (2 hours per study occasion + 3 hours for participant / surrogate information giving and consent) for 10 participants = 90 hours of translator time (face to face) To overcome language barriers for non-english speaking patients or surrogates, I have included a cost for interpreter services for around 10% of my target recruitment number (10 participants). This will be verbal face to face translation for patient study information, consent and qualitative data collection. The trust caters for a wide range of languages and making all documentation in different languages is not feasible.

	Year 1	Year 2	Year 3	Total Cost (£)
	1995	1995	1995	5985

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	8895	3565	4015	16475

Training And Development

Type of cost: NHS
Description: Applicant PhD Tuition Fees
Item Description: PhD Tuition fees
Justification of Cost: Brunel fees. I have added a 5% uplift to each year beyond year 1

	Year 1	Year 2	Year 3	Total Cost (£)
	4786	5025	5277	15088

Type of cost: NHS
Description: Short Courses
Item Description: Brunel Doctoral Modules - Year 1
Justification of Cost: Doctoral research: getting started; embedding EDI into research; academic resilience; critical thinking and writing.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Brunel Doctoral Modules - Year 2
Justification of Cost: ANOVAs in SPSS; delivering conference presentations; doctoral research: moving forward.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Brunel Doctoral Modules - Year 3
Justification of Cost: Doctoral research: completing successfully; preparing thesis submission.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Increasing Participation of Ethic Minorites in Health and Social Care Research
Justification of Cost: Increasing Participation of Ethnic Minorites in Health and Social Care Research (Centre for Ethnic Health Research / NIHR, eLearning): I recognise the need to increase my awareness of EDI in research; this training will help me achieve more diverse ethnic representation in my research. (online)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to qualitative interviews

Justification of Cost: Introduction to qualitative interviews (University of Oxford, 1 day). Practical and theoretical elements of qualitative research interviews – refreshes and builds on skills learnt on the Brunel qualitative research methods MSc module. First month of proposed start date. (online)

	Year 1	Year 2	Year 3	Total Cost (£)
	480	0	0	480

Type of cost: NHS
Description: Short Courses
Item Description: Qualitative data analysis: Theory and Practice
Justification of Cost: Qualitative data analysis: Theory and Practice (National Centre for Research Methods, 1 day): To develop my skills in qualitative data preparation, coding and identifying themes. (online)

	Year 1	Year 2	Year 3	Total Cost (£)
	140	0	0	140

Type of cost: NHS
Description: Short Courses
Item Description: Mixed Methods Design and Analysis
Justification of Cost: Design, sampling, analysis and integration of qualitative and quantitative methods, National Centre for Research Methods, 2 days, Manchester - cost includes travel and accommodation

	Year 1	Year 2	Year 3	Total Cost (£)
	862	0	0	862

Type of cost: NHS
Description: Workshops
Item Description: Inside a clinical trials unit
Justification of Cost: Inside a Clinical Trials Unit (Queen Mary University London Pragmatic Clinical Trials Unit, 9 months, 4 hours/month) This extensive course provides learning and mentorship in a live setting. I will gain experience in the design, conduct and governance of pragmatic clinical trials by observation and one-on-one tutor supervision (12+ hours). This will equip me with the skills to conduct the feasibility RCT in study 2, and further post-doctoral research This will take place over years 1 and 2

	Year 1	Year 2	Year 3	Total Cost (£)
	1600	1600	0	3200

Type of cost: NHS
Description: Short Courses
Item Description: Mixed methods analysis and integration for health research
Justification of Cost: Mixed methods analysis and integration for health research (Warwick Medical School, 4 days): Principles of mixed methods analysis: transformation, synthesis, integration, and analysis. Cost includes 4 nights accommodation and return train to Warwick

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2472	0	2472

Type of cost: NHS
Description: Workshops
Item Description: Supervised diaphragm ultrasound scans with Dr Hatem Soliman in Year 2
Justification of Cost: Supervised diaphragm ultrasound scans (three months, 4 hours/week): Supervised by Dr Soliman (Consultant Intensivist, Harefield Hospital). As an internationally renowned expert in ICU point of care ultrasound, Dr Soliman will develop my skills in diaphragm ultrasound in preparation for study 2.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Randomised controlled trials course: A guide to design, conduct, analysis, interpretation and reporting
Justification of Cost: Randomised controlled trials course: A guide to design, conduct, analysis, interpretation

and reporting (NIHR Oxford Biomedical Research Centre, 5 days): This course will provide me with a thorough grounding in RCT methodology and the skills to conduct a successful RCT. Cost includes 4 nights accommodation in Oxford, and return train ticket

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1893	0	1893

Type of cost: NHS
Description: Short Courses
Item Description: Writing and publishing qualitative research
Justification of Cost: Writing and publishing qualitative research (UCL, 1 day): This course will support me in describing and writing up qualitative data. (online)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	75	75

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to linear and logistic regression models (5 days) Bristol
Justification of Cost: Introduction to linear and logistic regression models (University of Bristol, 5 days). I have some experience in regression analysis; this course will help with more complex data analysis from study 2 and beyond. (online)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1100	1100

Type of cost: NHS
Description: Short Courses
Item Description: NIHR eLearning: Leadership development - Year 3
Justification of Cost: NIHR eLearning: Leadership development online course

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Workshops
Item Description: Clinical academic visit to David McWilliams at Coventry Hospital
Justification of Cost: Clinical academic learning visit which I will take each year of my fellowship. Yearly cost includes the cost of return train journey only. Accommodation with family nearby.

	Year 1	Year 2	Year 3	Total Cost (£)
	45	45	45	135

Type of cost: NHS
Description: Workshops
Item Description: Lane Fox visits - Year 2
Justification of Cost: Lane Fox, 1 day per week for 4 months, May-Aug, Year 2. Normal commuting distance. I will advance my clinical practice with multidisciplinary leaders in the field.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Overseas Research Visits
Item Description: Overseas visit to Professor Langer’s lab in Leuven, Belgium
Justification of Cost: Professor Langer’s lab conducts research in rehabilitation and inspiratory muscle training in critical care. He is an international expert on IMT in critical care, with >100 peer reviewed articles. I will visit for a research and clinical visit. Cost includes travel (flights) and 4 nights accommodation

	Year 1	Year 2	Year 3	Total Cost (£)
	574	0	0	574

Type of cost: NHS
Description: Short Courses
Item Description: Emerging clinical leaders, Kings fund (6 days)
Justification of Cost: Emerging clinical leaders (King's fund, 6 days). This course will help my understanding and subsequent development of my leadership style as a clinical academic. These skills will help improve team collaboration, wellbeing, and importantly, patient care.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2634	2634

Type of cost: NHS
Description: Short Courses
Item Description: Advanced course in acute respiratory failure and mechanical ventilation
Justification of Cost: Advanced course in acute respiratory failure and mechanical ventilation (European Society of Intensive Care Medicine, 2 days) Directly linked to my aspiration to be a leader in ventilator weaning. Covers lung/diaphragmatic ultrasound, assessment of respiratory drive and effort, strategies to protect the lung and diaphragm during mechanical ventilation. Cost includes flights to Barcelona and 3 night accommodation

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1029	1029

Type of cost: NHS
Description: Short Courses
Item Description: PPI training: Facilitation: how to get the most out for your PPI activities
Justification of Cost: Facilitation: how to get the most out for your PPI activities (UCL, 2 hours). Will improve my skills in identifying research priorities and designing protocols with PPI members.

	Year 1	Year 2	Year 3	Total Cost (£)
	30	0	0	30

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	8517	11035	10160	29712

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	111599	97159	97790	306548

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: Queen Mary University of London
Justification of Cost: Estates associated with statistician

	Year 1	Year 2	Year 3	Total Cost (£)
	1128	1128	1129	3385

Type of cost: Other Indirect Costs
Name of Higher Education Institute: Queen Mary University of London
Justification of Cost: Indirect costs associated with statistician

	Year 1	Year 2	Year 3	Total Cost (£)
	3386	3387	3387	10160

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	4514	4515	4516	13545

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	3611	3612	3613	10836

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	9254	10836	20090
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	297294	0	297294
Total Research Costs Requested From Funder	306548	10836	317384

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**74289**	**78733**	**79579**	**232601**
Travel-subsistence-and-conference-fees	**0**	**1560**	**1020**	**2580**
Equipment	**7899**	**0**	**0**	**7899**
Consumables	**9748**	**15**	**15**	**9778**
Patient-and-Public-Involvement	**2251**	**2251**	**3001**	**7503**
Other-Direct-Costs	**8895**	**3565**	**4015**	**16475**
Training And Development	**8517**	**11035**	**10160**	**29712**
HEI-Indirect-costs	**3611**	**3612**	**3613**	**10836**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**115210**	**100771**	**101403**	**317384**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)

NHS Support Costs	2820	1763	1762	6345.0

Justification:
NHS Support Costs: Support costs include the cost of patient consent and identification for the stage 1 work, and the stage 2 study (both intervention and control arms)

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	2820	1763	1762	6345

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS Treatment costs	144364	90228	90227	324819.0

Justification:
NHS Treatment costs: Includes the cost of standard care for these patients including, blood sample collection & processing, physical/daily examination (ward round), maintanence of intravenous medication, X-RAY.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Treatment Costs Total	144364	90228	90227	324819

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS Usual Treatment / Standard Care costs	144364	90228	90227	324819.0

Justification:
NHS Usual Treatment / Standard Care costs: Includes the cost of standard care for these patients including, blood sample collection & processing, physical/daily examination (ward round), maintanence of intravenous medication, X-RAY.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	144364	90228	90227	324819

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment	0	0	0	0	0

costs/(savings)					

	Total (£)
NHS Support Costs requested from networks	6345
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	6345

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	317384
Total NHS (Support and Excess Treatment) Costs	6345
Total Funding Required	323729

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - Figure 1 Study design and procedures.pdf) is included as an appendix within this file.

(Research Timetable - Research Timetable .pdf) is included as an appendix within this file.

(SoECAT Template - T Jenkins SoECAT_Diaphragm focussed, patient tolerable IMT.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Dr
Forename(s)	Vicky
Surname	MacBean
Position	Lecturer in Physiotherapy

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Professor
Forename(s)	Cherry
Surname	Kilbride
Position	Professor

Supervisor Participant 2

Title	Professor
Forename(s)	Michael
Surname	Polkey
Position	

Practice Supervisor(s)

Practice Support Participant 1

Title	Professor
Forename(s)	David
Surname	McWilliams
Position	Professor, Critical Care and Rehabilitation

Practice Support Participant 2

Title	Dr
Forename(s)	Hatem
Surname	Soliman Aboumarie
Position	Consultant in Cardiothoracic Intensive Care

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	
Forename(s)	Laura
Surname	Henderson
Position	

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Ann
Surname	Gallagher
Position	Head of Department

Head of Department or Senior Manager 2

Title	Dr
Forename(s)	Nana
Surname	Theodorou
Position	Associate Director of R&D

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. References.pdf
2. Figure 1 Study design and procedures.pdf
3. Research Timetable .pdf
4. T Jenkins SoECAT_Diaphragm focussed, patient tolerable IMT.xlsx

Appendix #1: References.pdf

References

1. Intensive care national audit and research centre: Case mix Programme Public Report 2022-23: CMP-Public-Report-2022-23-Summary-Statistics.pdf.pdf (icnarc.org) (assessed 31st May 2024)

2. Bellani, Giacomo et al. "Epidemiology, Patterns of Care, and Mortality for Patients with Acute Respiratory Distress Syndrome in Intensive Care Units in 50 Countries." *JAMA* vol. 315,8 (2016): 788-800.

3. Dres M, Dubé BP, Mayaux J, et al. Coexistence and Impact of Limb Muscle and Diaphragm Weakness at Time of Liberation from Mechanical Ventilation in Medical Intensive Care Unit Patients. *Am J Respir Crit Care Med*. 2017;195(1):57-66.

4. De Jonghe B, Bastuji-Garin S, Durand MC, et al. Respiratory weakness is associated with limb weakness and delayed weaning in critical illness. *Crit Care Med*. 2007;35(9):2007-2015.

5. Goligher EC, Dres M, Fan E, et al. Mechanical Ventilation-induced Diaphragm Atrophy Strongly Impacts Clinical Outcomes. *Am J Respir Crit Care Med*. 2018;197(2):204-213.

6. Damuth E, Mitchell JA, Bartock JL, Roberts BW, Trzeciak S. Long-term survival of critically ill patients treated with prolonged mechanical ventilation: a systematic review and meta-analysis. *Lancet Respir Med*. 2015;3(7):544-553.

7. Béduneau G, Pham T, Schortgen F, et al. Epidemiology of Weaning Outcome according to a New Definition. The WIND Study. *Am J Respir Crit Care Med*. 2017;195(6):772-783.

8. Supinski GS, Callahan LA. Diaphragm weakness in mechanically ventilated critically ill patients. *Crit Care*. 2013;17(3):R120.

9. Wilson ME, Barwise A, Heise KJ, et al. Long-Term Return to Functional Baseline After Mechanical Ventilation in the ICU. *Crit Care Med*. 2018;46(4):562-569.

10. Zilberberg MD, Nathanson BH, Ways J, Shorr AF. Characteristics, Hospital Course, and Outcomes of Patients Requiring Prolonged Acute Versus Short-Term Mechanical Ventilation in the United States, 2014-2018. *Crit Care Med*. 2020;48(11):1587-1594.

11. Hill AD, Fowler RA, Burns KE, Rose L, Pinto RL, Scales DC. Long-Term Outcomes and Health Care Utilization after Prolonged Mechanical Ventilation. *Ann Am Thorac Soc*. 2017;14(3):355-362.

12. Bureau C, Van Hollebeke M, Dres M. Managing respiratory muscle weakness during weaning from invasive ventilation. *Eur Respir Rev*. 2023;32(168):220205.

13. Bissett B, Leditschke IA, Green M, Marzano V, Collins S, Van Haren F. Inspiratory muscle training for intensive care patients: A multidisciplinary practical guide for clinicians. *Aust Crit Care*. 2019;32(3):249-255.

14. Vorona S, Sabatini U, Al-Maqbali S, et al. Inspiratory Muscle Rehabilitation in Critically Ill Adults. A Systematic Review and Meta-Analysis. *Ann Am Thorac Soc*. 2018;15(6):735-744.

15. Elkins M, Dentice R. Inspiratory muscle training facilitates weaning from mechanical ventilation among patients in the intensive care unit: a systematic review. *J Physiother*. 2015;61(3):125-134.

16. Ballesteros-Reviriego G, Arbillaga-Etxarri A, Martí JD. Inspiratory Muscle Training: Back to Basics Must be the First Step?. *Arch Bronconeumol*. 2024;60(5):267-268.

17. Cader SA, Vale RG, Castro JC, et al. Inspiratory muscle training improves maximal inspiratory pressure and may assist weaning in older intubated patients: a randomised trial. *J Physiother*. 2010;56(3):171-177.

18. Martin AD, Smith BK, Davenport PD, et al. Inspiratory muscle strength training improves weaning outcome in failure to wean patients: a randomized trial. *Crit Care*. 2011;15(2):R84

19. Condessa RL, Brauner JS, Saul AL, Baptista M, Silva AC, Vieira SR. Inspiratory muscle training did not accelerate weaning from mechanical ventilation but did improve tidal volume and maximal respiratory pressures: a randomised trial. *J Physiother*. 2013;59(2):101-107.

20. Sandoval Moreno LM, Casas Quiroga IC, Wilches Luna EC, García AF. Efficacy of respiratory muscle training in weaning of mechanical ventilation in patients with mechanical ventilation for 48hours or more: A Randomized Controlled Clinical Trial. *Med Intensiva (Engl Ed)*. 2019;43(2):79-89.

21. Bissett BM, Leditschke IA, Neeman T, et al. Does mechanical threshold inspiratory muscle training promote recovery and improve outcomes in patients who are ventilator-dependent in the intensive care unit? The IMPROVE randomised trial. *Aust Crit Care*. 2023;36(4):613-621.

22. Roceto Ratti LDS, Marques Tonella R, Castilho de Figueir do L, Bredda Saad IA, Eiras Falcão AL, Martins de Oliveira PP. Inspiratory Muscle Training Strategies in Tracheostomized Critically Ill Individuals. *Respir Care*. 2022;67(8):939-948.

23. Van Hollebeke M, Poddighe D, Hoffman M, et al. High and low-intensity inspiratory muscle training (IMT) have comparable effects in patients with weaning difficulties. *ERJ Open Res. 2024. 10: 62*

24. Kocjan J, Adamek M, Gzik-Zroska B, Czyżewski D, Rydel M. Network of breathing. Multifunctional role of the diaphragm: a review. *Adv Respir Med*. 2017;85(4):224-232.

25. Dres M, Dubé BP, Goligher E, et al. Usefulness of Parasternal Intercostal Muscle Ultrasound during Weaning from Mechanical Ventilation. *Anesthesiology*. 2020;132(5):1114-1125.

26. Parthasarathy S, Jubran A, Laghi F, Tobin MJ. Sternomastoid, rib cage, and expiratory muscle activity during weaning failure. *J Appl Physiol*. 2007;103(1):140-147.

27. Haugdahl HS, Storli SL, Meland B, Dybwik K, Romild U, Klepstad P. Underestimation of Patient Breathlessness by Nurses and Physicians during a Spontaneous Breathing Trial. *Am J Respir Crit Care Med*. 2015;192(12):1440-1448.

28. Demoule A, Decavele M, Antonelli M, et al. Dyspnoea in acutely ill mechanically ventilated adult patients: an ERS/ESICM statement. *Intensive Care Med*. 2024;50(2):159-180.

29. Van Hollebeke M, Poddighe D, Clerckx B, et al. High-Intensity Inspiratory Muscle Training Improves Scalene and Sternocleidomastoid Muscle Oxygenation Parameters in Patients With Weaning Difficulties: A Randomized Controlled Trial. *Front Physiol*. 2022;13:786575.

30. Shei RJ, Paris HL, Sogard AS, Mickleborough TD. Time to Move Beyond a "One-Size Fits All" Approach to Inspiratory Muscle Training. *Front Physiol*. 2022;12:766346.

31. Demoule A, Jung B, Prodanovic H, et al. Diaphragm dysfunction on admission to the intensive care unit. Prevalence, risk factors, and prognostic impact-a prospective study. *Am J Respir Crit Care Med*. 2013;188(2):213-219.

32. Jung B, Moury PH, Mahul M, et al. Diaphragmatic dysfunction in patients with ICU-acquired weakness and its impact on extubation failure. *Intensive Care Med*. 2016;42(5):853-861.

33. Hodgson CL, Udy AA, Bailey M, et al. The impact of disability in survivors of critical illness [published correction appears in Intensive Care Med. 2017 Sep 15;:]. *Intensive Care Med*. 2017;43(7):992-1001.

34. Kamdar BB, Suri R, Suchyta MR, et al. Return to work after critical illness: a systematic review and meta-analysis. *Thorax*. 2020;75(1):17-27.

35. Schmidt M, Banzett RB, Raux M, et al. Unrecognized suffering in the ICU: addressing dyspnea in mechanically ventilated patients. *Intensive Care Med*. 2014;40(1):1-10.

36. Davydow DS, Gifford JM, Desai SV, Needham DM, Bienvenu OJ. Posttraumatic stress disorder in general intensive care unit survivors: a systematic review. *Gen Hosp Psychiatry*. 2008;30(5):421-434.

37. Hearn E, Gosselink R, Freene N, Boden I, Green M, Bissett B. Inspiratory muscle training in intensive care unit patients: An international cross-sectional survey of physiotherapist practice. *Aust Crit Care*. 2022;35(5):527-534.

38. Dres M, Goligher EC, Dubé BP, et al. Diaphragm function and weaning from mechanical ventilation: an ultrasound and phrenic nerve stimulation clinical study. *Ann Intensive Care*. 2018;8(1):53.

39. Ward ME, Eidelman D, Stubbing DG, Bellemare F, Macklem PT. Respiratory sensation and pattern of respiratory muscle activation during diaphragm fatigue. *J Appl Physiol (1985)*. 1988;65(5):2181-2189

40. Dres M, Similowski T, Goligher EC, et al. Dyspnoea and respiratory muscle ultrasound to predict extubation failure. *Eur Respir J*. 2021;58(5):2100002.

41. Pozzi M, Rezoagli E, Bronco A, et al. Accessory and Expiratory Muscles Activation During Spontaneous Breathing Trial: A Physiological Study by Surface Electromyography. *Front Med (Lausanne)*. 2022;9:814219.

42. Skivington K, Matthews L, Simpson SA, et al. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. *BMJ*. 2021;374:n2061.

43. Creswell, J. W., & Plano Clark, V. L. Designing and Conducting Mixed Methods Research. Thousand Oaks, CA: Sage. 2006.

44. Bjerregaard Alrø A, Svenningsen H, Korvenius Nedergaard H, Irene Jensen H, Dreyer P. Cognitive impairment in intensive care unit patients: A qualitative exploration through observations and interviews. *Intensive Crit Care Nurs*. 2024;81:103611.

45. Jenkins TO, MacBean V, Poulsen MK, et al. The metabolic cost of inspiratory muscle training in mechanically ventilated patients in critical care. *Intensive Care Med Exp*. 2023;11(1):41.

46. Nickels M, Erwin K, McMurray G, et al. Feasibility, safety, and patient acceptability of electronic inspiratory muscle training in patients who require prolonged mechanical ventilation in the intensive care unit: A dual-centre observational study. *Aust Crit Care*. 2024;37(3):448-454.

47. Schmidt M, Kindler F, Gottfried SB, et al. Dyspnea and surface inspiratory electromyograms in mechanically ventilated patients. *Intensive Care Med*. 2013;39(8):1368-1376.

48. Jonkman AH, Warnaar RSP, Baccinelli W, et al. Analysis and applications of respiratory surface EMG: report of a round table meeting. *Crit Care*. 2024;28(1):2.

49. Silva Junior EFFD, Campos SL, Leite WS, et al. Surface electromyography signal processing and evaluation on respiratory muscles of critically ill patients: A systematic review. *PLoS One*. 2023;18(4):e0284911.

50. Zaga CJ, Freeman-Sanderson A, Happ MB, et al. Defining effective communication for critically ill patients with an artificial airway: An international multi-professional consensus. *Intensive Crit Care Nurs*. 2023;76:103393.

51. Ten Hoorn S, Elbers PW, Girbes AR, Tuinman PR. Communicating with conscious and mechanically ventilated critically ill patients: a systematic review. *Crit Care*. 2016;20(1):333.

52. Vincent JL. Communication in the ICU. *Intensive Care Med*. 1997;23(10):1093-1098.

53. Braun V, Clarke V. Thematic Analysis: A Practical Guide. London: Sage. 2021

54. Creswell, J. W. A Concise Introduction to Mixed Methods Research. 2nd Edition. London: Sage . 2022.

55. Fetters MD, Curry LA, Creswell JW. Achieving integration in mixed methods designs-principles and practices. *Health Serv Res*. 2013;48

56. Hayes JA, Black NA, Jenkinson C, et al. Outcome measures for adult critical care: a systematic review. *Health Technol Assess*. 2000;4(24):1-111

57. Granja C, Morujão E, Costa-Pereira A. Quality of life in acute respiratory distress syndrome survivors may be no worst than in other ICU survivors. *Intensive Care Med*. 2003;29(10):1744-1750.

58. Corner EJ, Soni N, Handy JM, Brett SJ. Construct validity of the Chelsea critical care physical assessment tool: an observational study of recovery from critical illness. *Crit Care*. 2014;18(2):R55

59. Hart N, Sylvester K, Ward S, Cramer D, Moxham J, Polkey MI. Evaluation of an inspiratory muscle trainer in healthy humans. *Respir Med*. 2001;95(6):526-531.

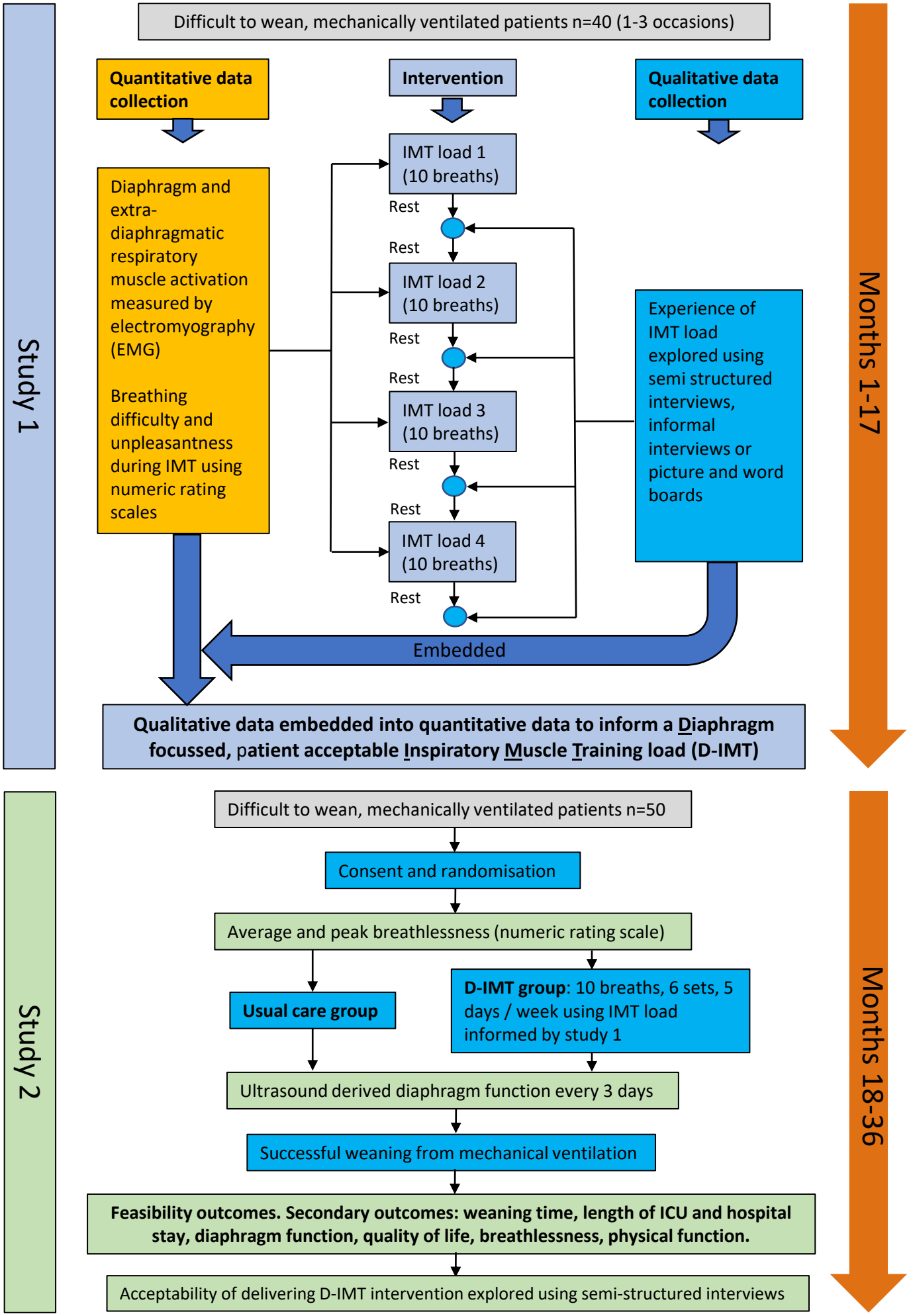
60. Supinski GS, Westgate P, Callahan LA. Correlation of maximal inspiratory pressure to transdiaphragmatic twitch pressure in intensive care unit patients. *Crit Care*. 2016;20:77.

61. Dubé BP, Dres M, Mayaux J, Demiri S, Similowski T, Demoule A. Ultrasound evaluation of diaphragm function in mechanically ventilated patients: comparison to phrenic stimulation and prognostic implications. *Thorax*. 2017;72(9):811-818.

62. Sim J, Lewis M. The size of a pilot study for a clinical trial should be calculated in relation to considerations of precision and efficiency. J Clin Epidemiol 2012;65:301-308

Appendix #2: Figure 1 Study design and procedures.pdf

Figure 1: Study design and procedures



Appendix #3: Research Timetable .pdf

Research Timetable

Year	Pre-DCAF	2025 - 2026												2026 - 2027												2027 - 2028											
Month		Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar
DCAF month		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Project management and governance																																					
Academic supervisor meetings **(monthly)**																																					
Practice supervisor meetings **(2 monthly)**																																					
Academic and practice supervisor combined meetings **(4 monthly)**																																					
PPI group meetings																																					
Statistician meetings																																					
Project steering group meetings																																					
Thesis preparation and writing																																					
Research project milestones																																					
Study 1																																					
R&D approval, ethics, project set up																																					
Data collection																																					
Data analysis																																					
Study 2																																					
R&D approval, ethics, project set up																																					
IMT procedure training for costed physiotherapists (4-8 physiotherapists)																																					
Data collection																																					
Data analysis																																					
Training and development																																					
Academic training																																					
Qualitative training																																					
Mixed methods training																																					
EDI training																																					
PPI training																																					
Clinical trials unit training programme **(4 hours per month)**																																					
Randomised controlled trials training																																					
Quantitative training																																					
Professional / clinical training																																					
Supervised diaphragm ultrasound scans **(4 hours per week)**																																					
Advanced course in acute respiratory failure and mechanical ventilation																																					
Secondment / research visit to Leuven, Belgium **(4 days)**																																					
Leadership training																																					
Secondment to Lane Fox Unit, London **(7.5 hours per week)**																																					
Secondment to Coventry University Hospital **(3 individual 37.5h weeks)**																																					
Conference attendance																																					
Dissemination and output																																					
Manuscript write up and submission																																					
PPI dissemination event																																					

Appendix #4: T Jenkins SoECAT_Diaphragm focussed, patient tolerable IMT.xlsx